
BLOCK 1

ENVIRONMENT AND SOCIETY **5**

BLOCK 2

ENVIRONMENT AND MOVEMENTS **95**

BLOCK 3

UNDERSTANDING: ENVIRONMENTAL ISSUES **161**

BLOCK 4

MEDIA COVERAGE OF ENVIRONMENT **233**

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Content

BLOCK 1	ENVIRONMENT AND SOCIETY	5
Unit 1	Introduction to Natural Environment	7
Unit 2	Introduction to Environmental Health	21
Unit 3	Historical and Social Perspectives in Environmental Problems	35
Unit 4	Democracy, Civil Society and Ecology	62
Unit 5	Environmental Ethics	84
BLOCK 2	ENVIRONMENT AND MOVEMENTS	95
Unit 6	International Environmental Policies, Agreements and Treaties	97
Unit 7	Environmental Sustainability	112
Unit 8	Environmentalism	131
Unit 9	Social and Civil Society Movements in India	146
BLOCK 3	UNDERSTANDING: ENVIRONMENTAL ISSUES	161
Unit 10	Introduction to Climate Change	163
Unit 11	Climate Change at the Global Level	181
Unit 12	India's Policy on Climate Change	202
Unit 13	Risk Society and Risk Communication	220
BLOCK 4	MEDIA COVERAGE OF ENVIRONMENT	233
Unit 14	Mass Media Representations of Climate Change	235
Unit 15	Language of Climate Change Communication	247
Unit 16	Economic and Political Drivers Behind Environmental Stories	261
Unit 17	Community Engagement and Green Media	277

BLOCK 1
ENVIRONMENT AND SOCIETY

THE PEOPLE'S
UNIVERSITY

BLOCK 1: ENVIRONMENT AND SOCIETY

Introduction

Unit 1 Introduction to Natural Environment: In this unit we occupy ourselves by defining the natural environment and linkages between people and natural environment by describing various segments of environment by explaining the human influence on environment and also the environmental sustainability.

Unit 2 Introduction to Environmental Health: Concept and scope of Environmental Health, Regional and global perspectives, Concept and requirements for healthy environment, Environmental quality, Human exposure and health impact, WHO, UN initiatives

Unit 3 Historical and Social Perspectives in Environmental Problems: Deals with Historical and Social Perspectives in environmental problems. The unit details on the limits to growth approaches to the study of the environment. The Marxian, Gandhian, Phule views have been discussed with reference to environmental problems, some environmental movements, technology, and society.

Unit 4 Democracy, Civil Society and Ecology: Deals with democracy, civil society, and ecology. The unit details on the historical development of science and technology, industrial revolution with special reference to India; technology adaptation: Third world countries and adaptation to technologies along with some case studies.

Unit 5 Environmental Ethics: Highlighted the Environmental Ethics relates to our moral obligation towards the environment. There are several approaches to environmental ethics viz., anthropocentrism, biocentrism and ecocentrism. Due to increasing apprehensions regarding environment and the growing awareness about the impact of human activities, there is a growing realization that environmental ethics has enormous scope. The environment is not something which we can remove oneself from. Human beings have started realizing that they have a right to healthy and safe environment. The unit explained the meaning of environmental ethic, environmental justice and the role of indigenous knowledge in protecting our ancient wisdom on environment. When a person realizes his environmental obligations he is able to think how an action or an event affects the entire range of environmental settings. For example, climate change which affects the world in various different ways. Further, the lifestyle of a person also impacts the environment and the future generations must be taken care of while consuming the limited resources present on the earth. The unit explains that it is not just the numbers but the ways of living of population which has a role in deteriorating the environment of a particular place.

UNIT 1 INTRODUCTION TO NATURAL ENVIRONMENT

Structure

- 1.1 Introduction
- 1.2 Learning Outcomes
- 1.3 Exploring Linkages Between People and Natural Environment
- 1.4 Understanding Our Environment
- 1.5 Segments of Environment
- 1.6 Atmosphere
 - 1.6.1 Chemical Composition
 - 1.6.2 Layers of the Atmosphere
 - 1.6.3 Function of Atmosphere
- 1.7 Hydrosphere
- 1.8 Lithosphere
 - 1.8.1 The Crust
 - 1.8.2 The Mantle
- 1.9 Biosphere
- 1.10 Ramifications of Anthropocene on Natural Environment
 - 1.10.1 The Human Dimensions of Global Environmental Change
 - 1.10.2 Carrying Capacity
 - 1.10.3 Loss of Ecosystem Services
 - 1.10.4 Environment and Health
- 1.11 Environmental Sustainability
- 1.12 Let Us Sum Up
- 1.13 Key words
- 1.14 Further Readings
- 1.15 Check Your Progress: Possible Answers

1.1 INTRODUCTION

We know that the magnitude, dimensions, and complexity of anthropogenic impacts on the natural environment are unprecedented. The developments in science and technology aid us in deciphering how environmental changes affect human well-being. Anthropogenic impacts on environment include both the direct and indirect effects. Nevertheless, technological powers exhibited by the humans cannot be overestimated.

At the beginning of modern era, the number of people in the world was probably about hundred million. In 1960, human population was about 3 billion. Our population has more than doubled in the last 40 years. The impact of the growing human population on the environment is quite unprecedented. For instance, fossil fuel burning pumps the GHGs into the

atmosphere and thereby affecting the human health. If the current human population growth persists, the human population could reach 9.4 billion by 2050. The unparalleled challenge is to cater the needs of growing population given the limitations of finite resources and inability of our technological prowess to minimize the ecological footprints. How then can the earth sustain this entire people and what is the maximum number of people that could live on the earth not just for a short time but over a long time?

We are aware of the fact that the natural environment has a significant impact on the quality of our life, health and sustainability of planet. Increasing population, urbanization and industrialization across the world places an enormous pressure on ecosystem, which affects the quality of air, water and food that we consume to maintain our life styles. New chemicals and industrial processes are being rapidly developed and sometimes produce unforeseen risks to human health and to the ecosystems.

1.2 LEARNING OUTCOMES

After completing this unit, you will be able to:

- define the natural environment and linkages between people and natural environment;
- describe the segments of the environment namely atmosphere, hydrosphere, lithosphere and biosphere; and
- explain the human influence on environment; and explain the environmental sustainability.

1.3 EXPLORING LINKAGES BETWEEN PEOPLE AND NATURAL ENVIRONMENT

People and nature are intimately integrated. We depend on nature for tangible resources and enviable services of nature without which our existence itself is impossible. To highlight a few services of our natural environment, effortlessly, we would be reminded of climate relating function of atmosphere, absorption of powerful UV radiations by stratospheric ozone and many ecosystem services rendered by our aquatic ecosystem. Human population explosion and associated urbanization and industrialization had impacted negatively the virtues of natural environment. Today, the discipline of environmental science is enlightening us about the interaction between people and nature and enable us to capitalize the beneficial interactions. As environment become more and more recognized as important, we become more earth centred. We accept that we have evolved on and with the earth and are not separate from it and we understand the need to celebrate our union with nature as we work towards sustainability. The dichotomy of the 20th century is giving way to a new idea that a sustainable environment and sustainable economy maybe compatible; people and nature intertwined and a success for one involve a success for the other. In this regard, environmental scientists are interested in determining the effect of pesticides on crops, in learning how acid rain affect vegetation, determining the impact of introducing an exotic game fish species into a pond etc. During the early

1960's environmental science became an active field of scientific investigation driven by both events and needs. The event that spurred environmental awareness includes the publication of Rachel Carson's silent spring (1962) along with other major environmental issues. The term environment and ecology are often used interchangeably but technically ecology refers only to the study of organism and their interaction with each other and environment. We damage the environment through the use, misuse and abuse of technology without fully understanding their long term effects on environment. The technological advances have provided us with nuclear power, the light energy, plastic, the internal combustion engine, air conditioning and refrigeration, genetic engineering, artificial intelligence, computer and scores of other advanced thing to live more comfortable. Many of these technologies, if not used properly affect the environment in ways we did not expect and in ways we may not be able to live with. Earth's natural resources and processes that sustain other species as well as humans are known as Natural capital. This includes air, water, soil, forest, grassland, wildlife, etc. Societies are the primary engines of resource use, converting material and energy into wealth, delivering goods and services and creating pollution. This provision of necessity and luxury is often conducted in ways that systematically degrade the earth natural capital, the ecosystem that support our life.

1.4 UNDERSTANDING OUR ENVIRONMENT

The origin of the word "Environment" is from the French word "Environ" which refers to "surrounding" or "to encircle". Environment can be defined as 1) the circumstances and conditions that surround an organism or group of organism 2) the social and cultural conditions that affect an individual or a community. Environment can also be defined as sum total of all external conditions and influences that affect the living organisms. Since human inhabits the natural world as well as the built or technological, social and cultural world, all constitutes important part of our surroundings. The environment comprises both biotic and abiotic factors. The 'natural environment' is made of living and non-living things (e.g., air, soil and water). Nevertheless, humans are integral part of the natural environment.

The concept of "environment" has evolved since it started to become a global issue in the early 1970s. At first, it was a kind of global recognition that the Earth's ecosystems are in fact fragile, and that human beings have been contributing much to its degeneration. When countries started to join efforts to strike a balance between improving the quality of human life and protecting the environment for the sake of future generations, a new awareness materialized. The social and economic welfare of human beings is closely linked to their environment. Any change in the socio-economic fields will have an impact on the earth's environment and vice versa, whether positively or negatively, immediately or eventually. And in many cases, negative results are irreversible. The Earth Summit held in Rio in 1992 concluded that the economic, social and environmental concerns are inescapably interlinked to development. Hence it pledged to eradicate environmental problems, reduce poverty and foster sustainable development

through integrated efforts and global cooperation.

Natural environment can be subdivided into two parts viz. the physical environment, and the biological environment. The physical environment comprises of lithosphere, hydrosphere and atmosphere. The biological environment is called biosphere (bios=life). Modern biosphere where men's activity dominates are sometime called as Anthrosphere or Neosphere. It includes besides natural environment complex of social, cultural and technological world that influences individual, community and ecosystem.

The biotic component of the environment interacts with the abiotic component of the environment leading unique ecosystems namely marine ecosystem, pond ecosystem, desert ecosystem, etc. As regards the living organisms of the ecosystem, they include mainly the producers, consumers, and decomposers. These organisms play a vital role in biogeochemical cycling of nutrients.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What do you understand by environment?

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2) Explain the linkages between natural environment and people.

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1.5 SEGMENTS OF ENVIRONMENT

There are four different segments of environment such as atmosphere, hydrosphere, lithosphere and biosphere.

1.6 ATMOSPHERE

Earth's atmosphere is unique. It is often described as thing veil, blanket of gases that surround the planet.

1.6.1 Chemical Composition

Atmospheric air is mixture of many gases that also holds many suspended liquid droplets and solid particles. Only two gases comprise about 99% of volume of air near the earth. The approximate composition of dry air, by volume at sea levels, Nitrogen 78%, oxygen 21% (necessary for life as we know it), argon 0.93%, and carbon dioxide 0.04%, together with very small amounts of numerous other constituents (see table 1.1).

Table 1.1: Composition of Air/Earth's Atmosphere

GAS	CHEMICAL/SYMBOL	VOLUME%
NITROGEN	N ₂	78.08
OXYGEN	O ₂	20.94
CARBON DIOXIDE	CO ₂	0.04
ARGON	AR	0.093
NEON	NE	0.0018
HELIUM	HE	0.0005
KRYPTON	KR	TRACE
XENON	XE	TRACE
OZONE	O ₃	0.00006
HYDROGEN	H ₂	0.00005

Source: De 2010

Because of the constant mixing by the winds and other weather factors, the percentage of gases in the atmosphere are normally constant. However, water vapour content is highly variable and depends on atmospheric conditions.

1.6.2 Layers of the Atmosphere

The atmosphere is composed of a series of four concentric layers: Troposphere, Stratosphere, Mesosphere and Thermosphere. These layers vary in altitude and temperature with latitude and seasons.

1.6.2.1 Troposphere

The troposphere extends to a height of approximately 12 km. The temperature of the troposphere decreases with increasing altitude by about 6 C° (11°F) for every kilometre. Weather including turbulent wind, storms and most clouds occurs in the troposphere. Tropopause is the top of the troposphere, which is a transition layer between Troposphere and Stratosphere.

1.6.2.2 Stratosphere

In the next layer of atmosphere, the stratosphere, there is a steady wind but no turbulence. The stratosphere extends from 12 km to 50 km above earth surface. Temperature is more or less uniform in this layer (-45 C° to 75 C°). Stratopause is the top of the stratosphere, which is a transition layer between Stratosphere and Mesosphere. It contains a layer of ozone critical to life because it absorbs sun's damaging ultra violet radiation. Ozone absorbs radiant energy from the sun and hence warmer temperatures are encountered in the stratosphere.

1.6.2.3 Mesosphere

Mesosphere is the layer of atmosphere directly above the stratosphere where temperature decreases with height, extends from 50km to 80km above the

earth surface. Temperature drops steadily in the mesosphere to the lowest in the atmosphere as low as -138 C°. Mesopause is the top of the mesosphere, which is a transition layer between Mesosphere and Thermosphere.

1.6.2.4 Thermosphere

The Thermosphere extends from 80km to 480km above earth surface and is very hot. Gases in the thin air of the atmosphere absorb X-rays and short wave ultraviolet radiation. This absorption drives the few molecules present to great speed, raising the temperatures in the process to 1000°C or more. The Thermosphere is important in long distance communications because it reflects outgoing radio waves back to earth without the aid of satellites.

1.6.3 Functions of Atmosphere

The atmosphere performs several ecologically important functions.

1. It protects Earth's surface from most of the ultra violet radiation, X rays and cosmic rays from space. Without the protective blanket of atmosphere, most life would cease to exist.
2. The atmosphere protects the earth from the high energy radiation, it allows visible light and some infra-red radiation to penetrate, and these warm the surface and lower atmosphere. This interaction between the atmosphere and the solar energy is responsible for the phenomenon of weather and climate.
3. It provides life sustaining oxygen, the macro-nutrient Nitrogen and Carbon dioxide needed for photosynthesis.

1.7 HYDROSPHERE

Hydrosphere encompasses water bodies namely oceans, streams, lakes, ponds, rivers, etc. This comprises all water resources both surface and ground water. Approximately, 326 million cubic miles of water cover much of the planet, but only 3% of this total water is fresh water. Three quarter of fresh water is found in polar icecaps and glaciers and unavailable for human use, and nearly all the remaining one quarter (ground water) is found under the earth crust in water bearing rock and or sand or gravel formations. Only a small proportion (about 0.05%) of all fresh water on earth is found in lakes, rivers, stream or the atmosphere. Obviously, this seems a small amount relative to Earth's total water supply. But it is more than enough, even this small amount if we keep free from pollution and distribute evenly, could provide for drinking, food preparation and agriculture needs of all people. We simply need to learn how to better manage and conserve fresh water available to us. Ground water and surface water are essentially one resource, both physically connected by the hydrological cycle. Ground water is created by the run-off surface water and that's why they are united as one.

1.8 LITHOSPHERE

The Earth is made up of a series of concentric rock zones namely crust, mantle and core. The upper part of the mantle to depths of about 100 km is

solid and together with the crust forms a relatively rigid unit known as lithosphere.

1.8.1 The Crust

Above the mantle is crust, which forms the outer skin of the earth. The thickness of crust ranges from 6km under the ocean to 70km under mountainous region.

Although the crust forms a very small part of the Earth as a whole, we gather practically all the resources to sustain our life.

Soil is the layer of bonded particles of sand, silt and clay that covers the land surface of the earth. Most soil develops multilayer. The top soils are the layer in which plant grows. This layer is actually an ecosystem composed of both biotic and abiotic components; inorganic chemicals, air, water, decaying organic material that provides vital nutrient for plant photosynthesis and living organisms. Below the topmost layers is the subsoil, which is much less productive, partly because it contains much less organic matter. Below this is parent material, the bedrock or the other geological material from which the soil is ultimately formed. The general rule of thumb is that it takes about 30 years to form one inch of topsoil to be formed from the parent material, the length of time depending on the nature of the underlying matter.

By mass the four most abundant elements in the earth are iron, oxygen, silicon and magnesium, which together accounts for approximately 93% of earth's crust. Nickel, sulphur, calcium and aluminium together make up another 6.5%. The remaining 0.5% or so of the earth's mass is made up of 84 naturally occurring elements.

1.8.2 The Mantle

The Earth's mantle, which lies between the core and the crust, is approximately 2900 km thick. The relatively thin upper part of the mantle is solid and rigid but the layer below it called as asthenosphere; although essentially solid, is able to flow extremely slowly.

1.9 BIOSPHERE

Biosphere can be construed as a life layer as it boasts of all the living organisms ranging from micro-organisms to macro-organisms. The biosphere extends from the bottom of the ocean approximately 11,000 meters below the surface, to the tops of the highest mountains about 9,000 meters above sea level. Although life exists throughout the biosphere, it is rare at the extremes, where conditions for survival are less than optimum. Most living things are concentrated in a narrow band extending from less than 200 meters below the surface of the ocean to about 6,000 meters above the sea level. The biological richness of biosphere is a function of factors like rainfall, temperature, etc.

The biosphere consists of all living organisms, energy and physical environments and there are continuous interactions between living and non-living organisms and physical environments and among the living organisms themselves. The biosphere is considered as an open system as there is

continuous flow of energy and output of matter. As long as the balance is maintained between the input of energy and matter and output of matter, the biosphere systems remains in equilibrium condition but when this critical balance is disturbed several ecological and environmental problems emerge. In natural state biospheric system is self sustaining and ecologically efficient, depending upon the intimate relationship between various components of biosphere and large scale cyclic mechanisms as this cycles influence the biotic, abiotic and the energy components of biosphere.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

3) What are the different segments of environment?

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4) What is the significance of atmosphere?

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1.10 RAMIFICATIONS OF ANTHROPOCENE ON NATURAL ENVIRONMENT

An unbounded growth is infeasible due to limited resources, and carrying capacity is a measure of this limitation. In 1798, Thomas Malthus proposed an exponential growth curve in his Essay on the Principle of Population, stating that a population with constant fecundity greater than its mortality will grow exponentially till it causes inevitable conflicts with the limited supply of resources. Diminishing resources and the expanding population will inevitably lead to scramble competition. Malthus's idea has further inspired Charles Darwin to adopt resource competition as one prerequisite of evolution via natural selection. Catton (1986) defined an environment's carrying capacity as its maximum persistently supportable load.

1.10.1 The Human Dimensions of Global Environmental Change

Present global environmental change can, to a large extent, be attributed to anthropogenic activities. Human interferences with the Earth's system are so significant that the recent era has been suitably named the "Anthropocene". Human activity continues to intensify sharply and increase pressures on the Earth's resources and, for instance, on the planet's capability to assimilate wastes. Rapid population growth and overpopulation has far reaching effect - ecological, economical and societal. The increasing population is putting greater burden on the earth natural resource base and environment.

The population growth may have positive or negative effects on the natural environment and wellbeing of human beings. Let us take for example the case of urbanization. A population shift toward urban areas means that a larger share of people will have access to health care, education, and other services; living standards are likely to improve. Greater population densities will enable more communities to capitalize on economies of scale. And concentrating population within an urban area can preserve adjacent natural habitat, assuming that urban sprawl is contained. At the same time, dense urban populations may produce more waste than the environment can absorb, leading to significant air and water pollution and a greater incidence of infectious and parasitic diseases. Cities often develop near fragile coastal areas or rivers or adjacent to fertile agricultural land. Rapid urban growth often takes over farmland, destroys wildlife habitats, and threatens sensitive ecosystems and inshore fisheries. Urban populations generally use more water for domestic and industrial purposes than rural populations.

1.10.2 Carrying Capacity

In ecology, carrying capacity is normally defined as the maximum population of a given species that can be supported indefinitely in a defined habitat without permanently impairing the human carrying capacity. Our planet is practically a closed system in terms of physical resources, and all human activity depends on these limited resources. Due to the increasing world population and the rapid diminishing of pristine ecosystems from our exploration for consumption, the concept of carrying capacity is essential and should be considered paramount in our future planning. The Club of Rome has warned us the possible consequences of world population growth and industrialization for food production and resource exhaustion.

The Earth's ecosystems and its human population are inherently connected. Earth sustains human activities by providing services like raw materials and energy. However, human activities tend to affect the natural resources and ecosystems. Environmental pollution adversely affect human health and wellbeing. An ecosystem consists of plants, animals (including humans) and micro-organisms that live in biological communities and which interact with each other and with the physical environment, enabling energy flow and biogeochemical cycling of nutrients. These interactions, for instance among species in a food web and their relation to biogeochemical cycling, are complex, non-linear, and contain lags and discontinuities, thresholds and limits. Consequently, ecosystems are considered complex, self-organizing systems, across both temporal and spatial scales. Ecological systems and their services play a fundamental role in supporting life on Earth at all hierarchical scales. For instance, they are essential in global material cycles like the carbon and water cycles.

1.10.3 Loss of Ecosystem Services

The Millennium Ecosystem Assessment 2001 defines “ecosystem services as the benefits people obtain from ecosystems” and classes services into four categories; Provisioning services provide goods such as food, freshwater, timber, and fibre for direct human use. Regulating services maintain a world

in which it is biophysically possible for people to live and provide benefits such as pollination of crops, water damage mitigation, and climate stabilization. Cultural services make the world a place in which people want to live; they include recreation as well as aesthetic, intellectual, and spiritual inspiration. Supporting services are the underlying ecosystem processes that produce the direct services described above, including the preservation of options; they include such services as nutrient cycles and oxygen production. Ecosystem services whether the provisioning or cultural kind matter and contribute to the well-being of humans. Food, fertile soils, clean water and recreation are just a few of the many services provided by nature to humans. The provision of these services, the reliability of supply, their quantity and quality, influence human security, basic material for a good life, health, good social relation and ultimately our freedom and choices, in short our well-being. Although people are often buffered from the natural environment by culture and technology, ultimately our livelihoods, health and even survival are completely dependant on ecosystem services and our access to them. Social systems and natural systems are inseparable and there is an increased use of terms such as socio-ecological systems. However, human action is changing many aspects of the earth's environments, its climate, land cover, oceans and the biochemistry of the fundamental cycles that sustain life and the diversity of life itself.

Approaching carrying capacity means the deterioration of ecosystem services, the loss of biodiversity and habitat heterogeneity and the break down of ecosystem resilience. This could lead to irreversible change, enlarged uncertainty to the biosphere, and even a regime shift of the world's living and climate systems as human population continues to expand, the planet's pristine ecosystems are degrading at a rapid rate due to man-made global environmental changes – climate change, habitat loss, biological invasion, soil erosion, and pollution, to name a few. Ongoing agricultural intensification, overfishing, industrialization, and urbanization further transfer how seminatural ecosystems function and behave, create many novel ecosystems that face high disturbance and gene flow. The planet's greatest asset – biodiversity, 100 million species strong – is quickly losing its grip. The rapid loss of species is estimated to be between 1000 and 10 000 times higher than the natural extinction rate, reaching 10 000 species extinction per year many of which disappear before they have even been formally described. Human society relies heavily on the health of ecosystem functions, the provision of ecosystem services and the benefit from biological diversity for cleaning air and water, stabilizing weather, maintaining soil fertility, dissipating waste, controlling pests, pollinating crops, generating power and discovering new antibodies, and providing food, timber, cloth, medicine, industrial material (coal, oil, gas, rubber, plastics, chemicals), together with minerals. We are not making anything new but utilizing products and services of the planet's ecosystem and physical system. To sustain humanity, we need to manage the planet's biosphere within its bearable margin to avoid disruptive regime shift and massive extinction; although some argue that we have already crossed the point of no return.

1.10.4 Environment and Health

Clean and hygienic environmental conditions are essential for health and wellbeing of humans. Common health impacts arise from air and water pollution and unhygienic conditions. Global environmental change which includes climate change, stratospheric ozone depletion, biodiversity loss, etc., have potential to affect human health. Health issues pertaining to air pollution include respiratory ailments, cardiovascular diseases; asthma and allergies, and other disorders. Since the interrelationship existing between environment and human health are complex, adoption of precautionary principle is widely accepted.

1.11 ENVIRONMENTAL SUSTAINABILITY

Sustainability is the capacity to endure. In ecology, the word describes how biological systems remain diverse and productive over time. For humans, it is the potential for long-term maintenance of well being, which in turn depends on the maintenance of the natural world and natural resources. Sustainability has become a wide-ranging term that can be applied to almost every facet of life on Earth, from local to a global scale and over various time periods. Sustainability is defined as a requirement of our generation to manage the resource base such that the average quality of life that we ensure ourselves can potentially be shared by all future generations. Development is sustainable if it involves a non decreasing average quality of life (Geir B. Asheim, "Sustainability," The World Bank, 1994). Long-lived and healthy wetlands and forests are examples of sustainable biological systems. Invisible chemical cycles redistribute water, oxygen, nitrogen and carbon through the world's living and non-living systems, and have sustained life since the beginning of time. As the earth's human population has increased, natural ecosystems have declined and changes in the balance of natural cycles have had a negative impact on both humans and other living systems. Paul Hawken has written that "Sustainability is about stabilizing the currently disruptive relationship between earth's two most complex systems; human culture and the living world. Evidence is strong that we are exceeding and eroding the earth's carrying capacity, that there are limits to growth on a finite planet. Effects are interactive, complex, unpredictable and escalating, as we head for a global average temperature rise of more than 2 degrees centigrade over pre-industrial levels. Sometimes 'environmental', 'social' and 'economic' are termed to be the three pillars of sustainability. Environmental sustainability is the context within which social and economic life happen.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) List down major human impacts on environment?

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- 2) What do you understand by environmental sustainability?

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1.12 LET US SUM UP

In this unit we have described the natural environment and linkages between nature and human being. We know that the Earth's environment is changing due to human activities. Undoubtedly, environmental change is occurring at all scales from local to regional to global level. The Global environmental change to a large extent is a product of human activities. These changes are to a large extent due to human activities, and they threaten the integrity and functioning of the ecosystems. Worse, future projected changes in the environment are likely to have even more severe consequences for safeguarding our natural environment and sustainable development.

1.13 KEY WORDS

Acid Rain: A rain or other forms of precipitation that is unusually acidic.

Climate Change: A change in weather over time and/or region; usually relating to changes in temperature, wind patterns and rainfall; although may be natural or anthropogenic, common discourse carries the assumption that climate change is anthropogenic.

Ecological Footprint : A measure of the area of biologically productive land and water needed to produce the resources and absorb the wastes of a population using the prevailing technology and resource management schemes; a measure of the consumption of renewable natural resources by a human population, be it that of a country, a region or the whole world given as the total area of productive land or sea required to produce all the crops, meat, seafood, wood and fibre it consumes, to sustain its energy consumption and to give space for its infrastructure.

Ecosystem Services: The role played by organisms, without charge, in creating a healthy environment for human beings, from production of oxygen to soil formation, maintenance of water quality and much more.

These services are now generally divided into four groups, supporting, provisioning, regulating and cultural services.

Ecosystem: A region in which the organisms and the physical environment form an interacting unit

Biosphere: The zone of air, land and water at the surface of the earth that is occupied by living organisms

Carrying Capacity: The maximum population that an ecosystem can sustain.

1.14 FURTHER READINGS

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1.15 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. “Environ” which means “surrounding” or “to encircle”. Environment can be defined as
 - 1) the circumstances and conditions that surround an organism or group of organism.
 - 2) the social and cultural conditions that affect an individual or a community. Environment can also be defined as sum total of all external conditions and influences that affect the living organisms.
2. Natural environment which is subdivided into two parts: (1) The physical environment, which includes non-living things and (2) the biological environment, which includes all life forms. Biological environment or people interact with the physical constituent of environment and modify the natural environment.

Check Your Progress: 2

1. There are four different segments of the environment such as atmosphere, hydrosphere, lithosphere and biosphere.
2. It protects Earth’s surface from most of the ultraviolet radiation, X rays and cosmic rays from space. Without the protective blanket of the atmosphere, most life would cease to exist. The atmosphere protects the earth from the high energy radiation, it allows visible light and some infra-red radiation to penetrates, and these warm the surface and lower atmosphere. This interaction between the atmosphere and solar energy is

responsible for the phenomenon of weather and climate. It provides life sustaining oxygen, the macro-nutrient Nitrogen and Carbon dioxide needed for photosynthesis.

Check Your Progress: 3

1. Sustainability is the capacity to endure. Sustainability is about stabilizing the currently disruptive relationship between earth's two most complex systems, human culture and the living world. Evidence is strong that we are exceeding and eroding the earth's carrying capacity, that there are limits to growth on a finite planet. Environmental 'social' and 'economic' are termed to be the three pillars of sustainability. Environmental sustainability is the context within which social and economic life happen.
2. Population produce more waste than the environment can absorb, leading to significant air and water pollution and a greater incidence of infectious and parasitic diseases. Rapid urban growth destroys wildlife habitats and threatens sensitive ecosystems and inshore fisheries. Noise is an emerging environment and health issue. Climate change, depletion of stratospheric ozone, loss of biodiversity, and land degradation can also affect human health. The major environment-related health concerns are related to outdoor and indoor air pollution, poor water quality, poor sanitation and hazardous chemicals. The related health impacts include respiratory and cardiovascular diseases; cancer, asthma and allergies, as well as reproductive and neurodevelopment disorders. Fine particulate matter and ground-level ozone are the main threats to human health from air pollution.

UNIT 2 INTRODUCTION TO ENVIRONMENTAL HEALTH

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 Concept and Scope of Environmental Health
- 2.3 Regional and Global Perspectives
- 2.4 Concept and Requirements for Healthy Environment
- 2.5 Environmental Quality
- 2.6 Human Exposure and Health Impact
- 2.7 Impact of Environmental Factors on Human Health
- 2.8 Let Us Sum Up
- 2.9 Key Words
- 2.10 Further Readings
- 2.11 Check Your Progress: Possible Answers

2.0 INTRODUCTION

Our environment and health are intricately linked to one another. A good quality environment has a positive impact on human health. When our environment is contaminated or polluted, then all life on this planet is affected. Environmental health includes the physical, chemical, and biological factors external to an individual that impacts human behaviours. Environmental health is very important as it helps in better understanding of relationships between our environment and the incidence of disease. It will help towards preventing disease and creating health-supportive environments. It also involves the measurement and control of the various environmental factors that may affect human health negatively. The impacts of environmental factors on health are varied and complex in nature. For example, environmental pollution can cause allergies, toxicity and even cancer. The linkages between exposure and disease are not that clear but there is a direct causal relationship in some cases. Urban air pollutants can cause respiratory diseases and heavy metals have been reported to cause neurological diseases. Environmental factors causing health disorders have not been given importance in policy decision and development planning activities. There are so many pathogenic biological agents and parasites that cause significant health problems. Further chemicals and hazards occurring at work environments are also some causes for diseases and deaths. Exposure to pollutants whether short term exposure or long term exposure causes diseases and in some cases chronic ailments. The environment may be polluted by natural phenomena like volcanoes or earth processes. But in the past decades especially after industrialization human beings have been responsible for such an increased and alarming level of pollutants in the environment. This is a cause of serious concern. It is only in the past two decades that there is an

alarm at the international levels with respect to the degradation of our environment and its consequences on human health and well-being. The United Nations (UN) and the World Health Organization (WHO) are together working to address the issues pertaining to health and environment. This is seen in culminating in a number of conferences and declarations.

This chapter brings out the relationships between environment, health and sustainable development.

2.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- define environmental health;
- explain the link between environment and human health;
- explain the importance of a healthy environment; and
- describe human exposure and health impacts.

2.2 CONCEPT AND SCOPE OF ENVIRONMENTAL HEALTH

Environmental health refers to components of human health and the quality of the environment. “It also refers to the theory and practice of assessing, correcting, controlling, and preventing those factors in the environment that can potentially affect the health of the present and future generations” (WHO, 1993). Environmental health aims to control disease and illness. It also aims to promote the well-being of humans by analysing the sources and hazardous agents in the environment. Environmental health services is defined as “those services which implement environmental health policies through monitoring and control activities” (WHO, 1993). This can be achieved by the improving our surroundings, maintain cleanliness and follow ecofriendly technologies. The professionals working in this discipline are known as environmental health officers or specialists, public health inspectors, and practitioners. Researchers and policy-makers also are needed in the field of environmental health. Health is now not only the responsibility of medical personnels, nursing and paramedical staff. It is every individual’s responsibility, the government, and NGO’s to work together and bring about the required change. Such changes can bring about good quality of the environment and affect the human health.

Let us now learn the definition of health as given by the World Health Organization (WHO).

Health: The WHO definition of health is that “it is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity” (WHO 1948). According to WHO, “the conditions required for health not only include the availability of resources to meet basic human needs and provide protection from all environmental hazards, but also require a sense of security and well-being”.

Environment and health are not separate entities. Both are driven by external

forces such as population and poverty. For a healthy nation the government should focus on the health of its people and maintain a positive balance between human activities and the environment. For achieving these goals the following are important. They include:

(a) Economic development must meet people's needs; and (b) Ecological sustainability must be achieved. Therefore joint action is needed at the local, national and global level by individuals and between governmental and nongovernmental agencies. Human activities can potentially damage the environment in many ways. Figure 2.1 will give you a better understanding of the relationship between health, environment and human activities. We can create healthy environments in a number of ways. This can be done by greening the environment and by managing wastes efficiently. Human activities like burning or emissions from industries can affect health, as toxic chemicals are released into the atmosphere polluting our physical environment.

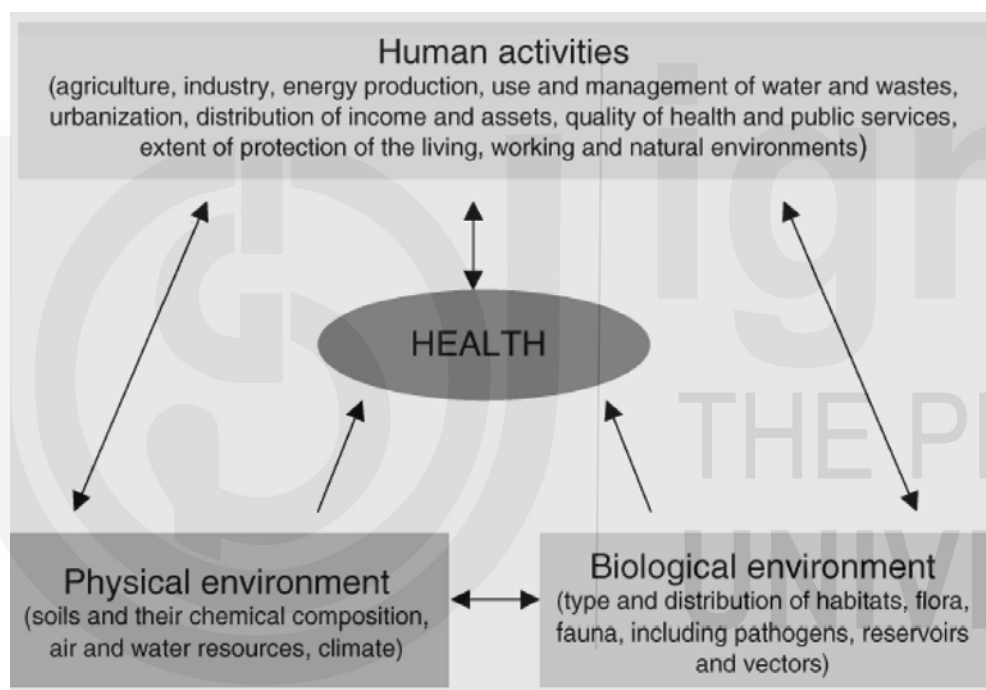


Figure 2.1: Interaction between human activities and the physical and biological environment (Source: Adapted from WHO (1992))

There are five basic disciplines in environmental health. They are environmental epidemiology, environmental toxicology, exposure science, environmental engineering, and environmental law. Environmental epidemiology deals with study of environmental exposures and human health. These include exposure to chemicals, radiation, biological agents, and so on. Toxicology deals with the study of how environmental exposures can lead to some disease or health issues. Generally toxicological experiments are carried out on animals in the laboratory to understand possible health outcomes in human beings. Exposure science studies the dose, quantity and exposure of a specific contaminant and its effects in humans. Such studies can also help in risk assessment. Environmental engineering deals with the application of engineering, technology and principles for the betterment of human health. Environmental law deals with the network of treaties, statutes

and regulations, for curbing the effects of anthropogenic activity on the environment. All the information obtained in the above mentioned disciplines can be used for risk assessment studies and how specific chemical exposure can cause significant risk to our health. This can be used for the implementation of suitable laws and environmental health policies. Technologies for minimising pollution impacts are also important for achieving the objectives of environmental health policy.

Activity

Now you know the WHO definition of environmental health. List some human activities that will affect health through their effect on the physical and biological environment.

2.3 REGIONAL AND GLOBAL PERSPECTIVES

Approximately half of the world's total population lives below the internationally defined poverty line. Out of this nearly 22% of the population in developing countries live on less than 1.25 US dollars per day. Poverty leads to a number of problems and to ill health. In 1900, pneumonia and influenza were the significant causes of death in the United States and other developed countries. In the 21st century diseases of the heart and other chronic diseases were the leading causes of death. Vaccines played an important role in reduction. Also improved nutrition, good housing, provision of clean water and safe disposal of wastes helped to reduce the communicable diseases. So the environment is directly linked to the health of an individual.

Presently environmental degradation is one of the major problems facing the world. It is caused due to population explosion, waste accumulation, pollution of the air, water and soil. Industrial wastes release a large amount of toxic substances which cause health impacts. Vehicular exhaust in cities is too high that the air quality is absolutely poor. Most of the polluted cities of the world are concentrated in developing countries. The world's oceans are also polluted with wastes which affects not only the quality of the water but also marine life. The number of known chemicals globally exceeds 14 million, of which over 60,000 are commonly used. All of these ultimately end up in the environment. Environmental problems occur at home, the workplace, the community level, and globally. Even though these problems may be viewed separately, they are all global issues affecting both local and remote populations. Hence, air pollution caused due to slash-and-burn practices in Sumatra severely affects the health of residents of Singapore and Malaysia. In the same way, the industrial pollutants released in northeastern United States cause acid rain, which adversely affects crops and people in the midwestern United States and southern Canada. The pollution of rivers upstream can adversely affect communities and countries downstream. Reports have shown that the children exposed to high levels of both primary pollutants and photochemical oxidants in urban air in Southern California have decreased lung function. In many developing countries, especially in China and India, the levels of pollutants observed are considerably higher than seen in developed countries. Indoor pollution is a problem in developing countries

where coal/charcoal is used for cooking in poorly ventilated houses. Approximately 40% of the world's population does not have access to clean drinking water. And almost 60% of the world's population does not have adequate facilities for waste disposal. Even in developed countries, water pollution is a serious issue as that of Milwaukee, Wisconsin. Here cryptosporidia contaminated the waters and caused causing severe illness and death. Presently developed countries are paying developing countries to accept their hazardous waste products. This is even more problematic as these developing countries have fewer resources to deal with the waste problem. So environment and health issues are major regional and global issues.

Global burden of environmentally related diseases

This is significant. Environmental factors on the burden of disease have been shown in approximately 21 percent of the overall burden of disease globally. Most of this burden falls on the developing countries. Figure 2.2 will help you to understand the global burden of environmentally related disease. About 28 percent of disease from diarrhoea occurs in the developing countries and a further 1.7 million children die from water contamination and diarrhoea. The next important health burden is acute respiratory disease which is due to poor indoor air quality.

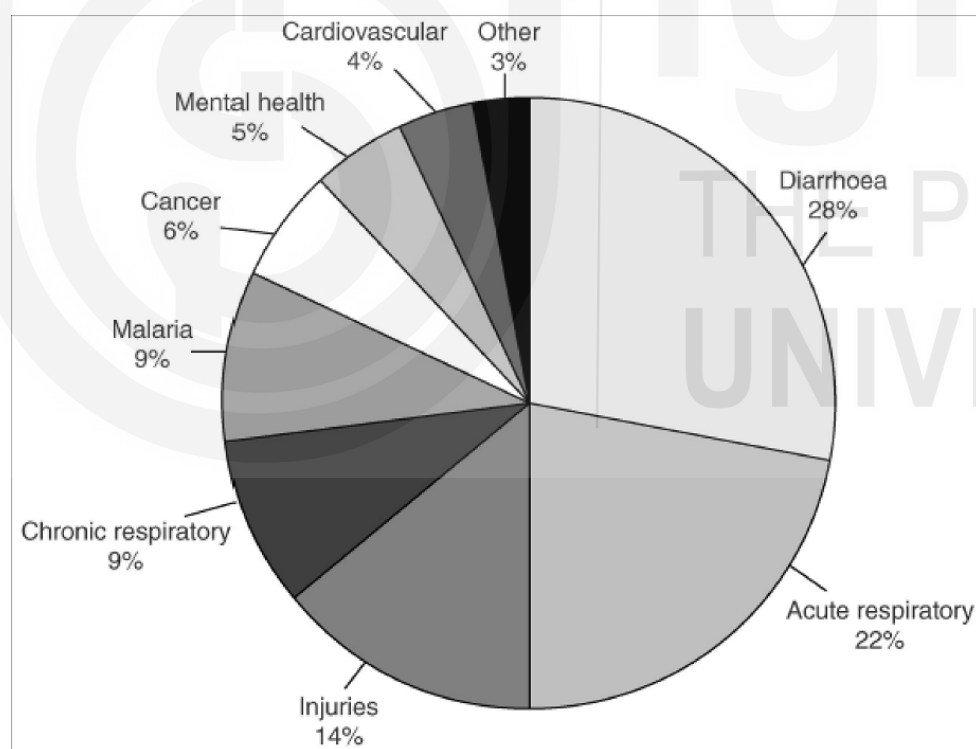


Figure 2.2: Global burden of environmentally related diseases

Source: Cairncross et al. 2003

Sustainable development: Health and environment

The World Commission on Environment and Development define sustainable development as “meeting the needs of the present generation without compromising the ability of future generations to meet their needs” (World Commission on Environment and Development, 1987). From time

immemorial human beings have interacted with the environment and have been using the available natural resources. But presently we are over utilizing these precious natural resources and we are facing the long-term consequences. Sustainable development addresses this imbalance and also addresses equity within the present generation. It has been observed that the poor are mostly exposed to environmental hazards. This also includes dangerous work places and inadequate access to safe food and water. Further, their houses are situated near industrial plants which are highly polluted. The importance of these aspects has been highlighted in the United Nations Universal Declaration of Human Rights: *'all people have the right to a standard of living adequate for the health and well-being of themselves and their family, including food, clothing, housing, health care and the necessary social services'* (UN 1948). It is only recently, that this declaration has been translated into action to ensure these rights for the present and future generations. A number of international meetings have been held such as the Earth Summit in Rio de Janeiro, Brazil, in 1992 where an integrated approach to the environment, health and sustainable development has been discussed. The first principle is that *'human beings are at the centre of concerns for sustainable development. They are entitled to a healthy and productive life in harmony with nature'* (UNCED 1992). The Earth Summit recognized that environment and human activities are often in conflict with each another. The summit emphasized nature should be respected and there should be a balance which has to be maintained between health and the environment.

Policy issues

From the above we understand that environmental degradation can lead to health problems. Therefore certain environmental policy interventions are required. The cost- benefit ratio for any such policy intervention will depend on the type of disease affecting the population and the exact state of the environment. These factors can vary between countries and also within countries. Some of the most important environmental issues in relation to human health include pollution and environmental degradation. The issue of air pollution and environmental quality calls for policies focusing on cleaner mechanisms of transport. In the same way policy interventions are also required for industries for reducing pollution impacts and associated health disorders. It is mandatory that whenever policies are formulated to address environmental problems, human health should be kept in priority.

Activity

Are there differences in the patterns of health and disease between developing and the developed? In your opinion what could be the reasons for this?

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. What is environmental health?

.....

- 2 Explain the basic principles of environmental health.

2.4 CONCEPT AND REQUIREMENTS FOR HEALTHY ENVIRONMENT

Healthy environments lead to healthy individuals. We have understood that environmental health is important to our health. A healthy environment encourages us to live in harmony with nature with an increased awareness of how our behaviour impacts our environment or surroundings. This includes promoting green initiatives and measures that improve the standard of living and quality of health in our environment. Therefore, maintaining a healthy environment is central for our healthy living. There are some important elements of the environment and their linkages to health. Now let us see some requirements for a healthy environment.

Good Outdoor and Indoor Air Quality: Poor air quality can lead to respiratory diseases and certain types of cancers. Access to clean air is extremely important. Presently air pollution is a major threat to human health. Air pollution is known to cause about 2 million premature deaths globally each year (WHO, 2008). Indoor air pollution is the 8th most important risk factor and responsible for 2.7% of the global burden of disease (WHO). Sufficient efforts have been made to reduce unhealthy air emissions. Reducing air pollution by use of clean fuels, green transport and reduced emissions from industries are solutions for clean environments.

Clean Water: The quality of the surface and ground water is extremely important. Access to safe drinking water is a requirement for healthy individuals. Almost 1.8 million people die every year from cholera, diarrhea and in this 90% are children under the age of five years 5, mostly in developing countries. Water that is contaminated with biological infectious agents or chemicals such as dyes can cause disease and illness. Therefore, the protection of water from contamination forms a fundamental aspect of environmental health. Reducing pollutant loads and treatment of effluents can help in healthy environments.

Reducing Toxic and Hazardous Wastes: Toxic and hazardous wastes from industries can affect the health of individuals. Long term effects cause irreversible damage to the human systems. Therefore, efforts should be made to reduce the use of hazardous substances and check their release into the environment.

Healthy Residences and offices: Some of these environments may expose us to different types of pollution. They include: Indoor air pollution, poor

sanitation, chemical hazards, biological hazards and physical hazards. All these hazards can impact human health and safety. Maintaining healthy homes, offices and communities is essential to environmental health. Many offices have started greening their campuses and also have made green roof tops. All this contributes to healthy environments.

Good Surveillance: Preventing exposure to environmental hazards also depends on the government health departments. Education and surveillance mechanisms are important monitoring for diseases, associated hazards, and educating the public. Some more methods are required to measure and address to the present environmental hazards for healthy environments.

Water quality: Water quality is an important parameter. Contaminated gives rise to various water borne diseases and some can be even fatal. By improving water quality and sanitation, health issues can be managed effectively.

Health and wellness are influenced by the places in which people live, learn, work, and play. Communities, including homes, schools, public spaces, and work sites, can be transformed to support well-being and make healthy choices easy and affordable. Healthy and safe community environments include those with clean air and water, affordable and secure housing, sustainable and economically vital neighborhoods.

2.5 ENVIRONMENTAL QUALITY

The quality of the environment literally means the quality of life. “Environmental quality is a measure of the condition of an environment relative to the requirements of one or more species and or to any human need or purpose” (Johnson et al. 1997). Environmental quality includes the natural environment as well as the built environment, such as air and water purity or pollution, noise and the potential effects which such characteristics may have on physical and mental health. We need to balance our future growth with the preservation and enhancement of our natural environment. With an increased demand for water and land resources it is important that when we address our future growth, we also take into consideration its environmental impacts. Some ways in which we can enhance our environmental quality include the following.

- Protect high-quality natural areas and sites with rare plants and animals
- Protect watersheds, wetlands, and streams to reduce pollution runoff, soil erosion, and flooding
- Provide clean, high-quality water to meet the domestic, economic, and recreational needs of the community
- Promote environmentally-sensitive development that respects natural areas and enhances the quality of our built environment
- Improve air quality
- Create a comprehensive and cost-effective solid waste management program that protects public health and the environment, promotes recycling, reduces

- the amount of solid waste, and successfully manages and reclaims landfill sites
- Enforce science-based and cost-effective environmental regulations
- Increase community understanding of and sensitivity to the area's natural systems and related environmental issues.

2.6 HUMAN EXPOSURE AND HEALTH IMPACT

Human beings are exposed to a wide variety of contaminants and pollutants in their homes and workplaces. The exposure to these pollutants can be short term or long term and the effects range from simple to complex diseases. Let us now see some pollutants humans are exposed to and their health impacts.

Air pollutant exposures and health effects: They can be caused due to natural phenomena and human activities. The most important sources of air pollution include inefficient modes of transport, vehicular exhaust, household fuels and waste burning, coal-fired power plants, and industrial activities. WHO air quality model confirms that 92% of the world's population lives in places where air quality levels exceed "WHO's Ambient Air quality guidelines" for annual mean of particulate matter with a diameter of less than 2.5 micro metres (PM_{2.5}). WHO guideline limits for annual mean of PM_{2.5} are 10 µg/m³ annual mean. PM_{2.5} includes pollutants such as sulfate, nitrates and black carbon, which go deep into the lungs and in the cardiovascular system, causing very high risk to human health. A large part of the urban population is exposed to excess of ambient quality levels of particulate matter, PM, nitrogen dioxide, benzene and ozone (EEA, 2002). Ozone and other photochemical oxidants are formed in the lower atmosphere by reactions of volatile organic compounds and nitrogen oxides in the presence of sunlight. Ozone can be transported over long distances and causes air pollution. High concentrations of ozone can result in higher incidences of respiratory disorders.

Radiation exposure and health Effects: Radiation can arise from different sources and it has sufficient energy to cause chemical changes in our cells and damage them even permanently. Some of the cells may die or become abnormal and can have altered states. They can damage the genetic material (DNA) and can cause cancer. High radiation exposure or acute exposure can cause sudden health effects and even death within hours or days.

Chemical exposure and health effects: These days we are exposed to a number of chemicals every day. It can be in the products we use or in food, water, the soil and the air. We can also be exposed to different chemicals at our workplace. Many of them may not be harmful. However, some chemicals can affect health negatively. Many products make life more efficient. But we should know how to use them properly and dispose them off in the required manner. Chemicals also give rise to several health disorders. The factors that lead to chemical toxicity include: the type, the quantity of chemical exposure, when and how long you are exposed, also the route of exposure, age of an individual and the general state of the individual. Sensitivity to certain chemicals can also differ from person to person. Children, pregnant women

and seniors may be at a higher risk. There can also be long term health effects from chemical exposures. Chemical toxicity can cause damages to our organs, weaken our immunity, bring about reproductive disorders, carcinogenicity and so on. WHO has established guideline values for more than 100 chemicals in drinking waters. High concentrations of nitrate in drinking water have serious health impacts. Nitrates can leach into ground waters from agricultural fields. Nitrates can be reduced to nitrite. It can cause methaemoglobinaemia, also known as the blue baby syndrome in babies. A report on the analysis of methaemoglobinaemia in Romania between 1985 and 1996 showed that 2, 913 cases were recorded and out of this 102 cases were fatal.

Noise exposure and health effects: High levels of noise can cause physical and psychological health disorders. High noise exposures in factories, industries can cause hearing loss, hypertension, heart diseases, sleep disturbance and annoyance. The amount of noise an individual is exposed to is important for the health effect to occur. In some cases long term exposures to high noise levels have caused changes in the immune system and also birth defects. Presbycusis or hearing loss normally occurs with age but high noise level exposure over a period of time can lead to presbycusis. Noise exposure has also been known to cause tinnitus (ringing sounds in the ears), vasoconstriction, and other cardiovascular adverse effects. Chronic noise exposure has also been associated with sleep disturbances and increased incidence of diabetes or high blood sugar levels.

Exposure to pesticides and their impacts: Dichlorodiphenyltrichloroethane (DDT) is an insecticide which is used in agricultural purposes. DDT was banned in 1972 in the United States of America, but certain countries still continue to use this chemical. DDT persists in the environment for a long time and also in animal tissues. It also bio accumulates in the tissues. We are exposed to DDT from foods, including meat, fish, and dairy products. DDT can be absorbed in the body by eating, breathing, or touching products contaminated with DDT. In the body, DDT is converted into several breakdown products called metabolites, such as: dichlorodiphenyldichloroethene (DDE). DDT and DDE are stored in the body's fatty tissues. In pregnant women, DDT and DDE can be passed to the foetus. Both chemicals are found in breast milk, resulting in exposure to nursing infants. When humans are exposed to DDT at high doses, symptoms like vomiting, tremors and seizures have been observed. Studies done on animals in the laboratories have shown deleterious effects on the liver and reproduction. DDT is considered a possible human carcinogen.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. What are some requirements for a healthy environment?

.....
.....

2. Describe some pollutants that humans are exposed to and their health impacts.

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.....

2.7 IMPACT OF ENVIRONMENTAL FACTORS ON HUMAN HEALTH

Our environment is directly linked to disease causation. The environmental impacts are presently a cause of serious concern. Environmental mutagens and carcinogens damage our cells, cause mutations and lead to cancer where the cells grow in an uncontrolled manner. Scientific data show that lymphomas, brain cancer, testicular cancer, prostate and breast cancer have increased dramatically over the past few decades. The environmental burden of diseases is comparatively higher in the developing countries when compared to the developed countries. Again in the developing countries the exposure to urban air pollution is affecting the health of a number of people. Climate change, population explosion and overcrowding, pollution are increasing in cities. All this impacts our health. Environmental factors can affect human health positively as well as negatively. Preventive medicine involves promoting positive environmental factors that sustain health. The negative environmental factors are threats to human health, and controlling them is known as public environmental health. Let us now discuss some environmental factors that affect our health.

Temperature and health: Temperature changes like cold and heat waves can impact human health. Studies have reported that unusual temperature fluctuations have resulted in cardiovascular, cerebrovascular and respiratory ailments. Also erythromelalgia condition is linked to rising temperatures.

Sunlight and health: Sunlight is very important for the synthesis of vitamin D in our body for healthy bones. However, sunlight can also be hazardous to human health. The ultraviolet rays from the sun can cause certain types of skin cancers and dermatitis. Ultraviolet B rays can cause rheumatoid arthritis in women.

pH of drinking water and health: The pH of drinking water whether acidic or alkaline can affect human health in a number of ways. When the water is acidic it can corrode metal plumbing lines. This can leach heavy metals such as lead, copper, zinc, manganese and iron. High levels of metals in drinking water such as lead can cause kidney disorders, neurological disorders, stroke and paralysis, high blood pressure and also cancers.

Humidity and health: Humidity levels in the indoor and outdoor environments can affect human health. Humid environments are highly conducive for the growth of microorganisms, molds, dust mites and mildew. This can cause allergies, respiratory problems, asthma and breathing difficulties. Low humidity levels can cause dry eyes, congestion, viral flu, and scaly skin.

Biotic environmental factors and health: Biotic factors like plants and animals cause infectious zoonotic and fungal diseases in human beings. Fungi, viruses and bacteria are biological agents that cause a number of infectious diseases.

Physical health effects can arise from natural events such as floods. Diseases such as communicable diseases, chest infections, coughs and colds, can also occur. Drought and desertification can also affect human health directly and indirectly. Other extreme weather events can lead to psychological disorders, disease or death, indirectly causing an increase in morbidity. Therefore environmental factors can also affect human health. Vulnerability and exposure, vary between different groups and areas. The elderly people and children are particularly at a higher risk. There is now a good understanding of the cause and effect relationships between pollution and human health. However, the health consequences of other environmental factors and exposures, such as chemicals present in the environment, are a result of complex interactions between the environment and humans. Every person is vulnerable to environmental impacts but the ability of people and societies to adapt and cope is different.

2.8 LET US SUM UP

This chapter introduced you to the fundamental concepts and scope of environmental health. The environment is polluted with contaminants from different sources such as industrial activities, domestic and agricultural activities. When humans are exposed to these contaminants on a short term or a long term basis we can be affected with diseases. It is in the past two decades that alarming levels of pollutants are observed in the environment which has called for international attention. In this unit we have understood that the environment is directly linked to human health. The impacts of the pollutants on human health are discussed. When human are exposed to chemicals or radiations especially from food, water and air various symptoms like cardiovascular diseases and deleterious effects on the liver and reproductive organs have been observed. Some chemicals like DDT is considered a possible human carcinogen. The importance of a healthy environment is discussed and how we can protect our ecosystems is also highlighted. Finally the importance of sustainable development is also discussed.

2.9 KEY WORDS

Sustainable development: It is defined as “meeting the needs of the present generation without compromising the ability of future generations to meet their needs” (World Commission on Environment and Development, 1987).

Dichlorodiphenyltrichloroethane: It is an insecticide which is used in agricultural purposes.

Health: The WHO definition of health is that “it is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity” (WHO 1948).

Environmental health: Environmental health comprises all aspects of human health and the environmental quality. This includes the physico-chemical, biological, and psychosocial parameters in the environment. “It also refers to the theory and practice of assessing, correcting, controlling, and preventing those factors in the environment that can potentially affect the health of the present and future generations” (WHO, 1993).

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2.11 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) Environmental health refers to components of human health and the quality of the environment. “It also refers to the theory and practice of assessing, correcting, controlling, and preventing those factors in the environment that can potentially affect the health of the present and future generations” (WHO, 1993). Environmental health aims to control disease and illness. It also aims to promote the well-being of humans by analysing the sources and hazardous agents in the environment. Environmental health services is defined as “those services which implement environmental health policies through monitoring and control activities” (WHO, 1993).
- 2) There are five basic disciplines in environmental health. They are environmental epidemiology, environmental toxicology, exposure science, environmental engineering, and environmental law. These disciplines contribute different information to describe problems and solutions in environmental health. Write each of these in detail.

Check Your Progress: 2

- 1) A healthy environment encourages us to live in harmony with nature with an increased awareness of how our behaviour impacts our environment or surroundings. This includes promoting green initiatives and measures that improve the standard of living and quality of health in our environment. Therefore, maintaining a healthy environment is central to increasing the quality of our life and years of healthy life. Some requirements for a healthy environment include: Good outdoor and indoor air quality, clean surface and ground Water, reducing toxic substances and hazardous wastes, healthy homes, offices and communities, good surveillance, good water quality and sanitation.
- 2) Human beings are exposed to a wide variety of contaminants and pollutants in their homes and workplaces. The exposure to these pollutants can be short term or long term and the effects range from simple to complex diseases. Some pollutants humans are exposed to and their health impacts are: Air pollutants, radiation exposures, chemical exposures, noise exposure and health effects, exposure to pesticides and their impacts.

UNIT 3 HISTORICAL AND SOCIAL PERSPECTIVES IN ENVIRONMENTAL PROBLEMS

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 Limits to Growth Approach to the Study of Environment
 - 3.2.1 Marxian Approach
 - 3.2.2 Gandhian Approach
 - 3.2.3 Phule approach
- 3.3 Development, Displacement, Relocation and Environmental Problems
- 3.4 Environment, Technology and Society
- 3.5 Let Us Sum Up
- 3.6 Keywords
- 3.7 Further Readings
- 3.8 Check Your Progress: Possible Answers

3.0 INTRODUCTION

Ever since the Industrial Revolution, there has been an increase in disruption of Earth systems leading to Environmental degradation. The decades 1960 and 1970 saw a gain in Environmental movements that demanded considerable changes in society. It was observed that there was a link between society and environment and that economic growth led to several environmental problems. This link was seen in books such as ‘Silent Spring’, ‘Limits to Growth’ and ‘Blueprint for Survival’. Many leaders and philosophers such as Karl Marx, Mahatma Gandhi and Jyotiba Phule stated their ideas for society and environment that bestowed a long-term impact. All of them have focused on community-centered approaches for sustainability.

Environmental problems such as global climate change, biodiversity loss, water pollution etc. have continued to worsen over years. Due to the extreme climate-related hazards, some areas worldwide are becoming less habitable. There is need of evidence of identified new places for carrying out the Resettlement and Rehabilitation activities along with significant policy and legislative changes in these systems. There is a growing need for holistic development of Environment, Technology and Society approach that helps people to connect one concept with the other. Technology and innovations should be such that there is minimal or no impact on the Environment while bringing maximum benefit to the society. Simultaneously, studying historical contexts helps us in appreciating cultural traditions and intellectual thoughts that have influenced environment and world at large. These contexts help us to derive conclusions from past and use them in present in developing sustainable strategies. Sustainable development, in its true sense, can only be

achieved by balancing the impact of economic activities, environment, health and well being of the community.

3.1 LEARNING OUTCOMES

After studying this unit, you will be able to:

- describe the importance of Environmental issues in the process of development especially for the future generation;
- identify some historical perspectives and ideologies of philosophers related to Environment;
- explain how developmental activities and Environmental events can result in displacement of people and get an insight into the policies related to resettlement; and
- discuss how Environment is related to Technology and Society.

3.2 LIMITS TO GROWTH APPROACH TO THE STUDY OF ENVIRONMENT

The genesis of Environmental History is usually seen through the Environmental movements taking shape in United States and Europe in 1960s and 1970s. The logical thinkers and philosophers of that time observed that there is a linkage between human activities and nature. It was predicted that human activities are changing the environment in a detrimental way that could make it difficult to survive on this planet. With rise of Industrial revolution, the eighteenth and nineteenth century had seen unhindered and uncontrolled usage of natural resources. Advances in science and technology provided humans with more power over their natural surroundings than before. There was a continuous race being carried out by the industrialized state to acquire more colonies. The period from the late 19th century till the end of the Second World War could be seen as one in which humans brought maximum damage to their natural environment. The beginnings of modern environmentalism was first said to be expressed in the path breaking and highly acclaimed work by Rachel Carson, who in her ‘Silent Spring’ and later works was highly critical of the industrial policies followed and reflected on the ill effects of human activities and rapid industrialization on the nature and natural habitats. As per the historical perspectives pertaining to Environmental issues, Economic activity was considered to be dependent on natural resources. The limitation of economic activity depended on capacity of resources. The ‘limit’ concept was introduced by some thinkers such as Malthus (1798), Ricardo (1817) and Marx (1867).

Definition

Donella and Dennis Meadows along with their team from Massachusetts Institute of Technology (MIT) titled their work/ report as ‘Limits to growth’ in 1972. A computer simulation was run through a software between economic and population growth and finite supply of resources. They predicted that there would be specific ecological limits if the consumption of resources followed the current trend. The limits applied to the economic growth and growth in production.

Conclusions of Report were:

- 1) As growth depends on consumption of resources, if the pace of industrialization, production and resource depletion continues at this level, the limits to growth may be reached within next hundred years. These limits to growth would result eventual decline in population and industrial capacities. It was stated that global industrial outputs, food and services would be limited within the next hundred years and thus the population would also reach a limit and then decline.
- 2) Maintaining stability between ecological and economic entities making use of sustainable practices may alter these growth trends. A capping for basic needs of a person should be designed so that each person is satisfied with the resources.
- 3) If people try to attain the balance between the ecological and economic entities and reduce the pace of industrialization, the greater could be the chances of success of coherent survival.

Criticisms of “Limits to Growth” approach

- 1) Technological input, which was supposedly an important component, was not considered and thus it was said that it does not predict technological advance.
- 2) The model used was a relatively simple computer model. One-dimensional variables were used to describe industrial output, resource depletion etc. that generally are more complex in nature. The approach was expected to be more quantitative.

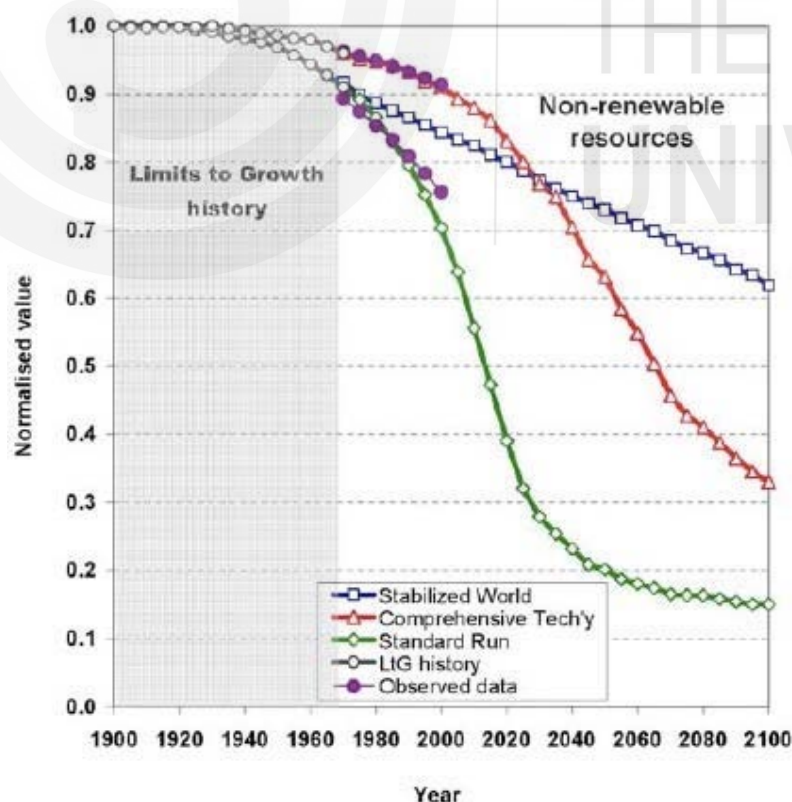


Fig. 3.1: Agreement of Limits to growth model with ‘business as usual’ scenario 30 years after report (Source: Earthbound report, 2010)

In the year 2008, the predictions of Limits to growth models were compared with the present scenario and it was found to relate well with present ‘business as usual’ conditions. The graph as depicted in **Fig. 3.1** shows that aggregate non-renewable resource depletion (shown with purple dots on red curve) agrees well with the standard run of Limits to growth model prediction (green curve). The depletion scenarios of certain resources match with the limits to growth model’s business as usual scenario. “Business as usual scenario” meant that pace of industrialization; population and resources consumed grew at the same rate as in the 1970 data that was used for predicting the future scenario.

Thus, we may observe that the Limits to Growth models agree well with the present scenario despite of certain flaws. The model was a stepping-stone for study regarding achieving sustainability.

Perspectives on Environment:

3.2.1 Marxian Approach



All progress in capitalistic agriculture is a progress in the art, not only of robbing the laborer, but of robbing the soil; all progress in increasing the fertility of the soil for a given time, is a progress towards ruining the lasting sources of that fertility
---Karl Marx, *Capital vol 1*

Karl Marx can be called as social scientist of the 19th century. He considered labor to be a process between man and nature. There was social and political unrest within the country and Marx highlighted the possibility of limited economic growth because of such unrest.

- Marx’s Theory is also known as “The Theory of Social Limit”
- Marx claimed that capitalism led to more production and rapid economic growth but could harm nature by depleting its resources. Marx also highlighted that laborers could be harmed with capitalism becoming increasingly productive, as it shall hamper their link with nature.
- According to Marx, Capitalism tended to decrease the wages to minimal levels. As population was continuously increasing, opportunity of work was reducing and laborers had to work for less wage.

Karl Marx’s ideas have left a mark on the entire world much more profoundly than any other philosopher. Marx introduced the concept of “alienated labor” in his early manuscripts. The alienated labor meant that people were being separated from nature due to the capitalist framework of society. Due to the industrial class capturing the common land, the rural people had no option of meeting their daily needs. The capitalists exploited their labor power. Rural people were also deprived of their spiritual needs associated with nature. The theme of his early manuscripts has been of the concept that exploitation of workers and nature go hand in hand. He stated that:

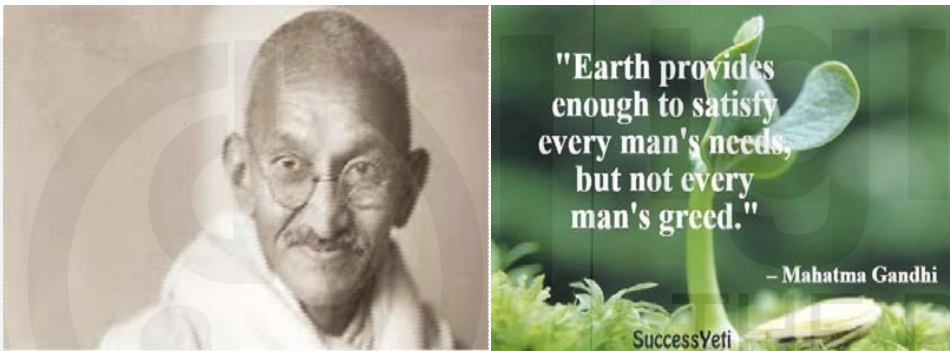
Man lives on nature – means that nature is his body, with which he must remain in continuous interchange if he is not to die. That man’s physical and

spiritual life is linked to nature means simply that nature is linked to itself, for man is a part of nature.

He also reiterated that people need to understand the privatized/capitalized society and how it impacts the nature if environmental problems are to be addressed. In 1860's, due to industrialization, there was division of towns in the country, which led to loss of soil fertility and its degradation. The people of those areas did not understand this phenomenon but there was a mention of this in Marx's work. He had mentioned about the imposition of burden of pollution and diseases due to industrialization.

Present ecologists who showed a belief in Marxian philosophy argue that capitalism leads to unsustainability and we need to shift towards greener practices in order to achieve sustainability. In order to transform society and shift towards sustainability, Marx's emphasis on amiable relationship with nature can be considered. Through this, material as well as spiritual needs from nature can be met.

3.2.2 Gandhian Approach



Mahatma Gandhi being a hard-core conserver of nature and environment inspired many environmental movements in India through his sayings and works. Some of his beliefs and works pertaining to environment were:

- **Integration:** Ecology is a subject that involves identifying and studying the relationship between biotic components and their environment. Human beings are dependent on environment for their daily needs.

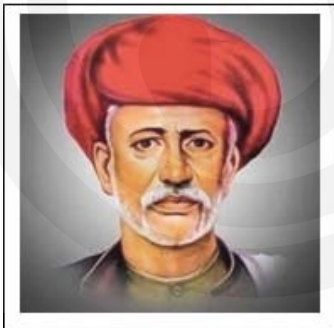
Gandhi had a belief in advaita (non-duality) and unity of man and all lives. Gandhi saw integration between various entities of nature and advocated for bringing in balance amongst all. He said that humans tend to have compartmentalization within their bodies where one part does not know what other is doing but nature treats every individual in the same manner. All living beings are treated equally by nature.
- **Wise use of resources:** According to Gandhi, a man should keep limited number of material needs and **have** plain living. He said that comfort is necessary but not to an extent that it becomes greed. Plain living and high thinking of mankind was his ideology of living.
- **Thoughts against Industrialization:** The thoughts of industrialization in the west disturbed him. He was somehow aware regarding the perils of industrialization. He said that India should never take industrialization as seriously as the west. During those times, England was ruling over

other kingdoms and stripping the world like locusts. Gandhi, however, followed few Western thinkers, who cherished and praised the pre-industrial era. He followed John Ruskin whose thoughts were that industrialization had destroyed the harmony between man and nature.

- **Purification through nature:** Gandhiji's views were influenced by Jainism. The foundation of this influence was Jainism's inherent character of considering nature as living entity. He firmly believed that human beings should respect all forms of nature as their equals and must consider them as their brothers and sisters.
- **Emphasis on limited violence to nature:** He considered subjecting nature to limited violence and persuaded people to just take those resources from nature that were necessary for their sustenance. He emphasized that exploiting nature for short-term receipt of resources was not a good practice and should be minimized. One such instance that has been described often is when his close disciple had a habit of breaking an entire twig of neem tree just for his need of four or five neem leaves. Gandhiji termed that as violence and asked his disciple to seek an apology from the tree for his wasteful and wrong deed.
- **Influence in Environmental movements and Environmentalists:** Gandhian thoughts on nature were a motive for carrying out different environmental movements in India. One amongst the largest environmental movements in India, Chipko movement, derived inspiration from Gandhi's philosophy.
- **Influence on Environmentalists:** The philosophy to concentrate and include rural population in addressing environmental concerns can be attributed to Gandhiji. Many environmentalists and people working for society such as Ramchandra Guha and Madhav Gadgil have credited Gandhi for their work. The common feature amongst all of their work is the special focus given over to work in rural areas, which also has been derived from Gandhian thoughts. Similarly, Vandana Shiva, an environmental activist acknowledged Gandhian influence while starting the movement of "Seed satyagraha" to protect biodiversity. This movement enabled farmers to understand their seed rights. Besides this, campaigns were also conducted to boycott unjust Intellectual Property rights
- **Nature cure:** Gandhi had said that resources are enough in nature for everyone provided people take just the amount that is required by them. He even requested the medical practitioners of that time to cure people using medicinal plants so as to harness traditional knowledge. He also insisted on indigenous ways of treatment that had inclination towards nature.
- **Emphasis on artisan skills:** He focused on village-based technologies such as Charkha. He focused on the fact that human labor should not be replaced indiscriminately and artisan tools should be given their due importance. His viewpoint was that excessive use of technological equipments was not a real sign of progress and technology should be used in unison with indigenous methods of manufacturing.

- **Consideration of nature as sacred:** Gandhi considered living in co-existence with non-human world. His cottage in Sevagram faced several events of presence of snakes or other animals. But he advised people not to kill the snakes. Rather, he encouraged them to pick up snakes and release them at far off places in their natural habitat. He considered all non-human entities as sacred and divine and considered it important to live in harmony with them.
- **Conservation:** Gandhi always focused on wise use of resources. Besides conserving money and other personal resources, he even ensured that only an optimal amount of water be given to plants. He conveyed the idea of conserving resources for the future generation of the country thus focusing on sustainability.
- **Stress on Environmental problems:** The patterns of growth and development which were genesis of industrialization were considered a disease by Gandhiji. According to him, all mess created by industries such as pollution, climate change or biodiversity loss needed to be cured by focusing on indigenous practices.
- **Focus on Cleanliness:** Mahatma Gandhi awakened the Indian community on improving their public image on sanitation while underlining the importance of keeping the surroundings clean. He has been a true inspiration behind Swachh Bharat scheme launched on 2nd October, 2014.

3.2.3 Phule Approach



Though Women are great enablers in the field of environment and sustainable development, their contribution has been undermined for centuries in India. They being the chief resource manager, manage everything from water, fuel and food to other resources, while simultaneously taking equal responsibility in agriculture. Asia's largest environmental movements such as Chipko Movement and Green belt movement has

highlighted results when women undertake the responsibility to contribute in sustainable development. Some benefits that the society can reap by educating women are demonstrated through Figure. 3.2.

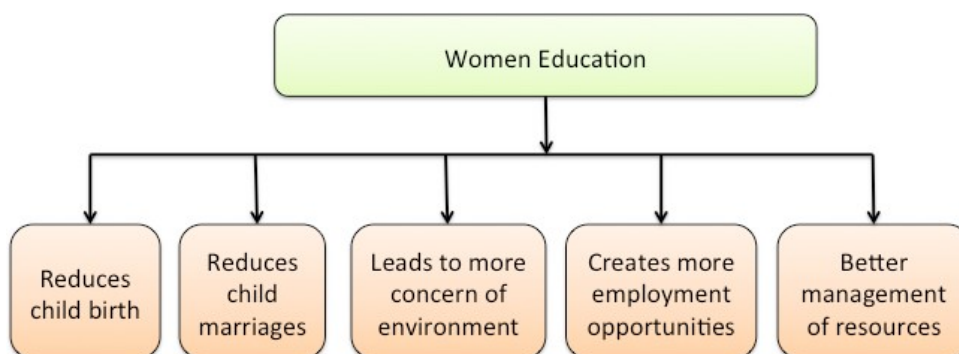


Figure 3.2: Benefit of Women education to society and environment

Thus, for an overall development of the society, it is imperative to teach and enlighten women so that they can further educate their children and encourage them to differentiate between right and wrong.

Jyotirao Govindrao Phule, also known as Jyotiba Phule recognized this and worked diligently for Women empowerment and improvement and enlightenment of economically and socially weaker sections. He understood the implicit relation that uplifted women could bring about in strengthening of the society, in both social and environmental terms. Some of his contributions have been in the varied fields of:

- **Improvement of social status of women:** During 1860's and 1870's, women did not hold a significant social status and were frequently ill treated by their own families. The prevalent caste system ensured that one particular caste was deprived of all kinds of rights in society and education. In his struggle for emancipation of Women and eradication of caste system, Jyotiba Phule opened the first school at Pune meant to educate families from lower caste. Besides facing a stiff opposition from upper caste Brahmins, he and his wife, Savitribai, continued with the tireless effort to ensure education for all, irrespective of their caste and creed.
- **Female education and training:** Mahatma Phule knew that widening the horizons for women shall widen the philosophy and opinions people carry, thus further awakening the society. He thought that emancipation of the oppressed classes was possible only through social reforms and education. His institutions for underprivileged females were a modest start of a larger public movement and sign of societal transformation. Jyotiba was extremely passionate about bringing this societal transformation through teaching females, while keeping his wife Savitribai in fore front as a role model. He stated that the improvement, which comes about in a kid because of a mother's larger involvement in decision-making, is significant for a better psyche of the child. He emphasized that futuristic and holistic development of a nation demands a non-oppressed female fraternity. He called attention to the fact, that those who are actually worried about the well being as well as welfare of the nation should take note of the state of women and make every attempt to impart knowledge to them in case they sincerely wish the nation to improve.

With this particular idea they initially began the school for women. Most people weren't prepared to send their kids to school but Lahuji Ragh, Raut Mang and Ranba Mahar convinced the communities regarding the positive aspects of women getting educated.

- **Equality of society:** Mahatma Phule was a sign of the revolution of societal equality. His vision promulgated equality as a force for making social, national and economic change all over India. The obsolete socio-religious structure defined by caste system bred inequality. Both Savitribai and Jyotirao battled for gender equality and worked their way against prejudiced caste system in spite of huge criticism by the Brahminic and Manuwadi forces. Jyotiba highlighted that prosperity of a

country is ascertained only when women take a lead role and are treated equal to men.

- **Child marriage and widow remarriage:** To be able to empower females he opposed Child marriage. Simultaneously, he advocated for widow remarriage and arranged for an “Ashram” for widows. In those times, widow remarriage was a taboo amongst Hindus. A lot of widows were young and not all of them would live in a fashion in which the orthodox individuals expected them to live. Several of the widows resorted to abortion or left their illegitimate children to the fate of theirs by leaving them on the streets. So, he established a home for newborn infants of widows in 1863 to avoid the infanticides.
- **Education and Progress:** Phule stated that six kind of progress were feasible just with education. In the introduction to the guide Shetkaryacha Asud, written by him, he elucidated these six philosophies as: Without training wisdom was lost; Without wisdom morals have been lost; Without morals growth was lost; Without development wealth was lost; Without wealth the underprivileged had been ruined. Thus, he gave a brand new meaning and perspective to the concept of education
- **Fight against Untouchability:** He worked as a pioneer of liberating Dalits from oppression and tyranny. Although the renaissance had started but still Dalits were humiliated and discouraged to take part in the social life or functions. Their presence in education related sectors were abysmally low. They weren't permitted to make use of public places, visit temples or take drinking water out of public wells & tanks. Phule made highly effective pleas to the society to abolish “untouchability” and revamp the caste system. He revolted against the unjust caste system which rendered a large group of people “without any rights” for centuries altogether.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) State the conclusions of ‘Limits to Growth’ report.

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2) State some of Mahatma Gandhi’s perspectives on the Environment.

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3.3 DEVELOPMENT, DISPLACEMENT, RELOCATION AND ENVIRONMENTAL PROBLEMS

Relocation of humans has been a phenomenon that has been practiced for long for different reasons. Some people moved due to economic factors such as employment while others moved due to environmental events such as floods, droughts or famines. These migrations were either for a limited time period or were more permanent in nature.

Generally, government agencies pursue programs for population relocation either due to looming hazards, threats of violence or developmental activities. While the first two causes are more unforeseen, the relocation for the development projects is more systematic in nature. Developmental projects require large land area and there is need to displace local people. Accordingly, local people may have to relocate as per the government norms and policies.

Definitions:

The following set of definitions is given by International Organization for Migration (IOM) which is the inter-governmental organization that works in the field of migration. IOM became a subsidiary of United Nations in September, 2016:

Migrant: A person who has moved across an international border or within a state away from his/her habitual place of residence.

People are sometimes forced to move to a different place due to some factors known as push factors. On the other hand, if people are attracted to a different place due to better avenues available, such factors are known as pull factors. Factors that can affect a migrant are indicated in the following Table 3.1:

Table 3.1: Push and Pull factors due to which migration occurs (Adapted from Jaccob, 2013)

Push-factors Countries of Origin	Migrants	Pull-factors Countries of destination
➤ Population growth, young age structure ➤ Inadequate educational institutions, medicare and social security	Demographic Factors and Social infrastructure	➤ Stable population, population decline, demographic ageing ➤ Welfare state benefits, educational institutions, medicare, social security
➤ Unemployment, low wages ➤ Poverty, low consumption and living standard	Economic factors	➤ Labour demand, high wages ➤ Welfare, high consumption and living standard

➤ Dictatorships, shadow democracy, bad governance, political upheaval ➤ Conflict, (civil) war, terrorism, human rights violation, oppression of minorities	Political factors	➤ Democracy, rule of law, pluralism, political stability ➤ Peace, security, protection of human and civil rights, protection of minorities
➤ Ecologic disaster, desertification, lack of natural resources, water shortage, soil erosion, lack of environmental policy	Ecological factors	➤ Better environment, environmental policy, protection of natural resources and environmental protection
➤ Decisions of the family or the clan ➤ Information flows, media, transferred picture of country of origin ➤ Possibilities of (ir) regular immigration, routes of trafficking	Migrant flows and migrant stocks	➤ Diaspora, ethnic community ➤ Information flows, media, transferred picture of destination country ➤ Possibilities of (ir) regular immigration (right of residence)

Refugee or Asylum seekers: A person who flees across an international border due to a well-founded fear of being ill-treated due to his race, nationality or political opinion.

Internally displaced person: A person who is forced to depart from their home, but has not crossed an international border.

Environment migrant: A person who moves either temporarily or permanently due to some sudden or progressive environmental changes such as volcanoes, hurricanes etc. They may move either within their territory or abroad.

Internal environmental migrants: A person who voluntarily relocates domestically for reasons of environmental changes.

Environmental refugee: A person who because of difficult environmental conditions, is forced to depart from home and cross international borders.

Environmental Internally Displaced Person: A person who is forced to leave home but remains within the state of which he/she is a national.

These definitions constitute legal classifications for people for their entitlements towards certain relief/ benefits. But there are certain limitations associated with these definitions. However, it is more on the jurisdiction of the agency to provide relief based on state's rule. For example, if any agency volunteers have to provide relief to refugees, they may/may not consider

providing relief to environmentally displaced persons. Thus, sometimes these definitions bring about a sense of confusion during entitlements.

Rehabilitation and Development

Resettlement process is generally immune to the emotional or social aspects of migrating people. Migrants may have certain emotional attachments with the previous dwelling sites and may want the new site to be similar to earlier ones. A good rehabilitation process should take care of this while simultaneously ensuring a smooth transition between the below mentioned factors:

- Landlessness to land-based resettlement
- Joblessness to re-employment
- Food insecurity to safe nutrition
- Homelessness to home reconstruction
- High morbidity and mortality to improved health and well being and
- Social disarticulation and deprivation of common property resources to community reconstruction and social inclusion

Thus, this kind of phased rehabilitation ascertains that a citizen is granted the same freedom and rights that he/ she was enjoying before relocation.

Environmental Problems leading to Migration

Amongst many problems that Environmental changes pose, few are categorized as follows:

1) Global Warming

- Global warming is a major environmental problem caused by release of green house gases like carbon dioxide, methane, etc.
- Significant rise in global temperatures due to the trapping of solar radiations by green house gases can result in further increase in temperature which is a phenomenon related to climate change. Climate change may lead to melting of polar ice fields that leads to rise in sea level.
- This melting of glaciers could result insubmergence of low-lying lands as a result of rising sea level. This will lead to inundation of large part of land thereby affecting a great mass of people.
- Another major impact of global warming would be on the rainfall patterns of different parts of the world. It is estimated through climate change models that there would be large increase in precipitation near equator, Arctic and Antarctic region. Also, the drier regions are estimated to become even drier (CarbonBrief report, 2018).
- As climate change impacts are rising, number of people moving internally to different cities or across border is also rising. It is estimated that about 25-300 million people are likely to be displaced due to the impacts of climate change by 2050 (Stern and Stern, 2007).

The international frameworks and achievements pertaining to human mobility with regard to climate change are depicted in Table 3.2.

**Table 3.2: International frameworks addressing human mobility
(Adapted from Wilkinson et al, 2016)**

Year	Achievement
2001	Preparation and implementation of National Adaptation Programmes of Action.
2005	Global Commission on International Migration delivered final report.
2006	Global Migration Group established
2008	Migration first mentioned in Poznań COP14
2010	Cancun COP16 established Cancun Adaptation Framework, Advisory Group on Climate Change and Migration and National Adaptation Plans Process.
2012	Nansen Initiative launched
March 2015	Sendai Framework for Disaster Risk Reduction 2015-2030 adopted
September, 2015	2030 Agenda for Sustainable Development adopted.
October, 2015	Endorsement of the Protection agenda
November, 2015	Intended Nationally Determined Contributions submitted ahead of Paris COP21
December, 2015	Task Force on displacement established in the Paris Agreement, Paris COP21
May, 2016	Platform on Disaster displacement launched
September, 2016	Adoption of New York Declaration for refugees and migrants
October, 2016	New Urban Agenda adopted

Migration occurring because of climate change was first mentioned in United Nations Framework Convention on Climate Change (UNFCCC) documents in year 2008 in Poznań Conference of Parties (COP14). However, no clear outcome of such migration was elucidated in the conclusions presented for COP14. The concept of Climate induced migration was revisited in Paris agreement COP21. But for some reason, the terminologies have been kept ambiguous and there is no clear understanding of it, either to stakeholders or policy makers. With the increasing wrath of climate change being faced by the world, there is an essential need to define and collate the various perspectives of migration under one framework.

2) Land degradation

- Land degradation refers to temporary or permanent decline in productive capacity of the land and soil, which is used for production of food and fodder (Scherr and Yadav, 1996).

- Deforestation and desertification, chemical contamination, pollution, overgrazing, soil erosion etc. are some factors that contribute to land degradation (Source: TAFE Queensland and Department of Natural Resources Module ASP3056: Environmental Impacts:Issues, 1999).
- These factors that lead to land degradation may result in alteration of physical/ chemical properties of soil, loss of biodiversity and increased hazards to humans occupying these degrading lands.
- Over 1.3 billion lives are impacted globally due to land degradation (UNDP Regional Service Centre for Africa, 2017).
- Populations that are affected by land degradation tend to lack access to land area as well as resources. The limited resources, thereby, leads to more conflicts amongst these populations (BOGOTA: Thomson Reuters Foundation Report, 2018).
- A need was felt world wide, for some agency to take charge of issues of migration resulting due to land degradation. International Organization for migration (IOM) which became a subsidiary to United Nations in 2016 now addresses issues related to migration resulting from land degradation, drought and desertification.

3) Industrial Accidents

- Industrial accidents such as toxic spills from mining/non-mining activities, oil spills or transport accidents can have severe impacts on health of affected people and environment.
- There may be losses such as body injuries, economic loss, property damage or environmental damage such as material/ resource loss and pollution of air, water or soil.
- Bhopal Gas tragedy of India that occurred in December 1984 was one of the major industrial disasters in India (ENVIS-NIOH Newsletter, 2014). There were about 3787 deaths and several permanent-disabling injuries that occurred because of methyl isocyanate gas release. People suffered from severe eye irritation, burning in respiratory tract, vomiting and stomach pains. A major population living in surrounding areas had to be extricated away from the plant.
- It is difficult to prepare for the occurrence of such accidents as they are very different from natural hazards and one cannot be certain as to how to handle such scenarios.

4) Natural disasters

- Natural disasters are catastrophic events, majorly caused by nature, that cause harm to communities and loss to economy. In grave circumstances, a need for outside assistance to displace the affected community to a new place arises (Petrucci, 2012).
- Some of these disasters are floods, droughts, hurricanes, volcanic eruptions, earthquakes etc (And now viruses too...).

- Loss of resources, shelter and security generally observed after a disaster may lead to several migrations especially in developing countries.

Other such events that lead to Environmental displacements are depicted in Figure 3.3.

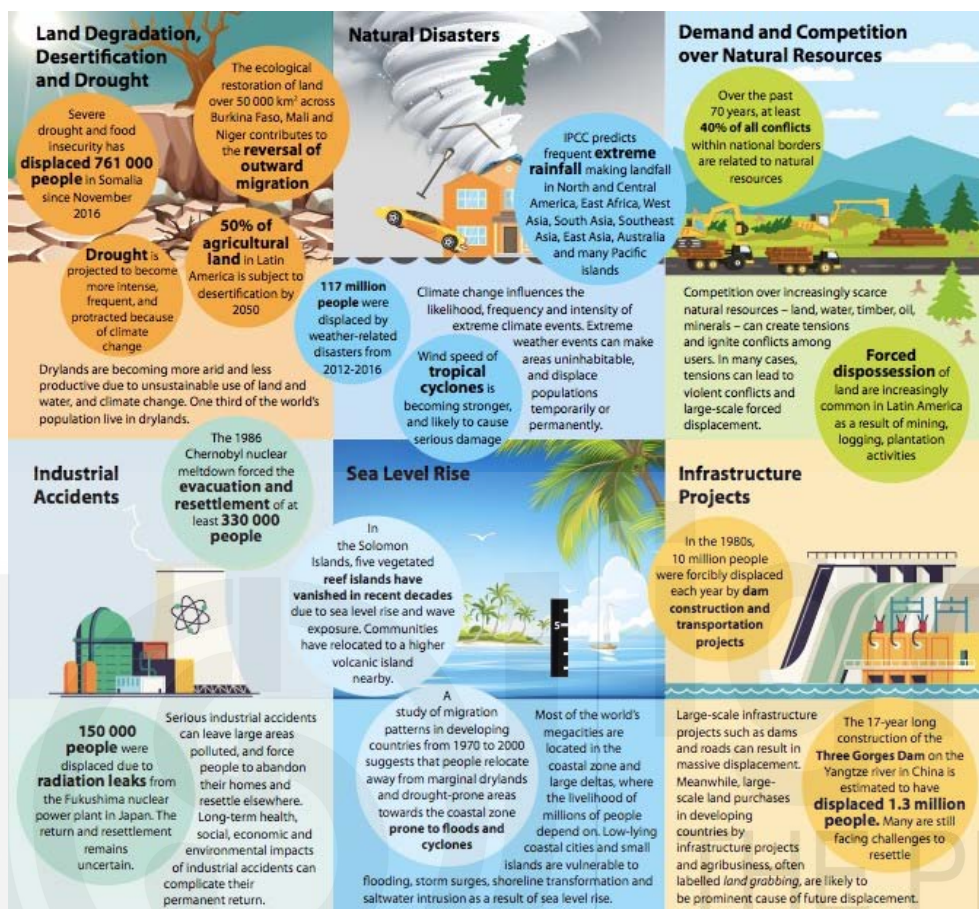


Figure 3.3: Environmental Problems leading to displacements
(adapted from Brown, 2017)

Impacts of Environmental migration

- Migration can prove to be a favorable adaptive strategy if governments and legal institutions provide residency rights, education and employment opportunities to the aggrieved parties. If right conditions are set up, it can contribute to social and economic development. Although in most cases, the outcomes of migration come with their own associated costs.
- Migrants are prone to environmental hazards, meager access to health services and education. Generally, they are also exploited in employment.
- Migration also takes a toll on the mental health of individuals. They undergo various stressful conditions in the recipient country as they face language problems and disruption of their social network. Frustration and loneliness is a common phenomenon amongst these individuals.
- The migrants may sometimes carry certain infectious diseases, which

may be harmful for the host country. In other cases the migrants may be healthy and catch diseases from host country. Thus, there are several healthcare challenges that are faced by the migrants and host countries.

- The health impacts of environmental migration are depicted in Figure 3.4.

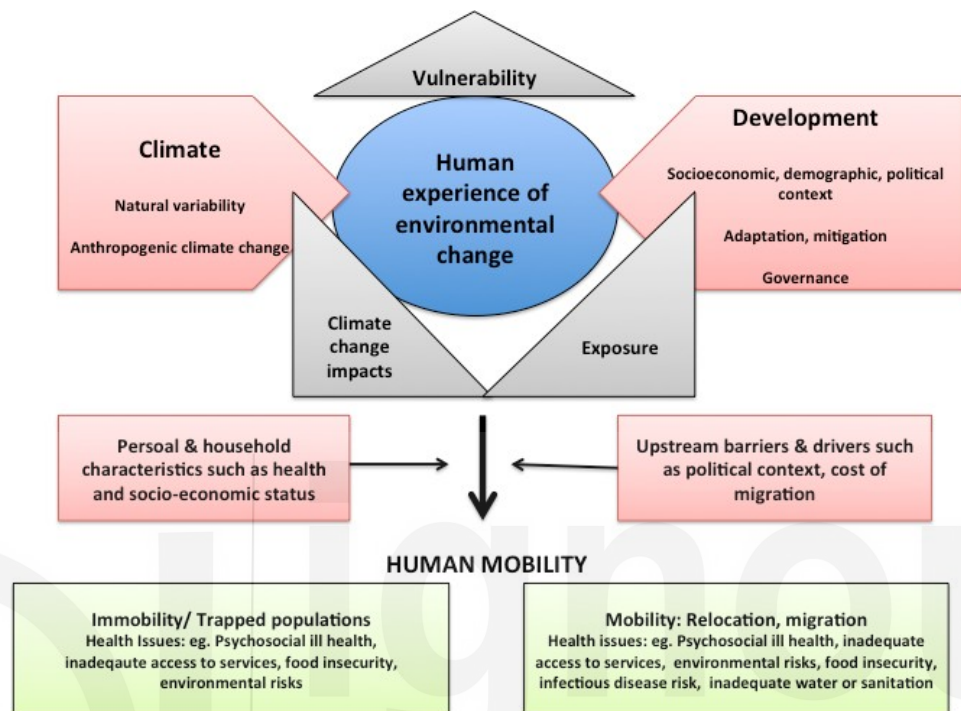


Figure 3.4: Health impacts of Environmental migration
(Adapted from Schwerdtle et al, 2018)

- Thus, the solution for all these problems lies in proper planning and reduction in vulnerability levels of the migrant population through enhanced livelihoods.

Policy Analysis

Rehabilitation and resettlement (R&R) process/ policies should primarily aim at helping the displaced persons who lose their lands due to developmental activities, conflicts or natural calamities. In most of the cases, displaced people try to go back to their original dwelling sites as and when the situation normalizes as in case of natural calamities. But, in developmental activities, they may have to resettle to a new place permanently. Most countries have their own R&R policies. Based on the type of developmental project, there are different organizations in each country that take the responsibility of providing basic amenities to such people. World Bank and Asian Development Bank are two of such organizations that sponsor development projects and bear the cost for resettling people.

The history of development projects in India dates back to post independence. There was no specific R&R policy defined for displaced persons and state governments used to pass orders for the affected communities randomly. Different states promulgated different policies based on type of projects going on in that state. Though Land acquisition act had a standing, displaced persons had to satisfy with a piece of land and their

livelihood was not guaranteed. The first for malpolicy related to R&R came up in the year 1985 when the World Bank refused to help government for Sardar Sarovar Dam project. But this policy was meant majorly for R&R of Tribal communities and other communities did not benefit from it. In 1993, the Ministry of Rural Development prepared another draft of such policy but that too was criticized. The formal Policy on R&R that addressed many earlier deficiencies was framed in 2007 and came to be known as National R&R Policy of India.

The overview of policies related to rehabilitation and resettlement can be observed in Figure 3.5.

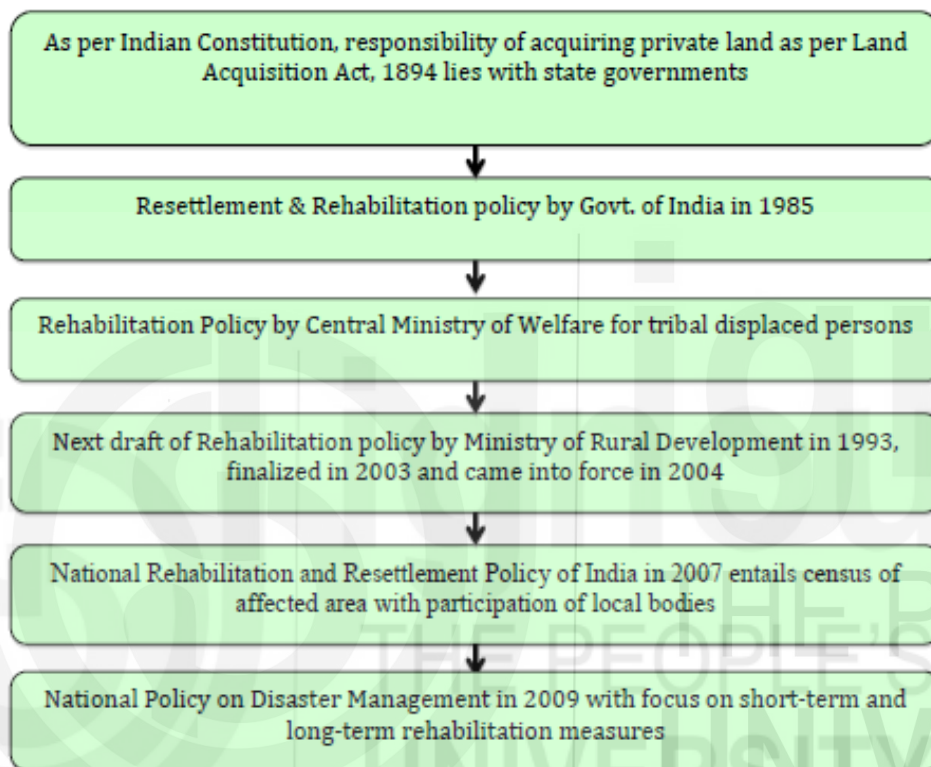


Figure 3.5: Policies related to Rehabilitation and Resettlement in India

As per the National Rehabilitation and Resettlement Policy (2007) of India, the R&R benefits include land for land, compensation packages, rehabilitation grant that varies according to the type of land and purpose of acquisition, employment for at least one person per nuclear family, vocational training, scholarships, skill development opportunities, allotment of outsources contracts shops/other economic opportunities and providing labor work at the project site.

Further, the STs and SCs are accorded separate R&R benefits, aiming to safeguard their ethnic, linguistic and cultural identity, ensuring that they be located in similar ecological settings as their earlier place of dwelling. The policy also has provisions for monitoring and reviewing the implementation of the R&R plan and also a mechanism for grievance redressal. This redressal is either the responsibility of the Administrator for Resettlement and Rehabilitation or of a designated R&R Committee. It is also mandated that there should be an R&R Committee at the state and district level. The issues

of grievance redressal are looked after by an Ombudsman appointed by the state concerned.

However, the National R&R policy lacks in the area of covering the disaster-induced displacement. This has been seen as a major drawback. In case of Disaster induced- displacements, there is a need to look into the policies related to disaster in India.

The Disaster Management Act(2005) mandates the Ministries and departments at all levels to prepare a disaster management plan. They have the responsibility of providing relief in case of any disasters and take care of rehabilitation and reconstruction operations if required. The financial resources needed for carrying out the above-mentioned activities is to be borne by National Disaster Response Fund (NDRF). If mitigation measures need to be taken, the expenses can be released from the National Mitigation Fund. The Act does lack a definite plan for resettlement and rehabilitation activities and thus displaced people are sometimes in distress.

The above mentioned flaws have been duly taken care of in The National Policy on Disaster Management (2009). This policy has focused on both short-term relief operations as well as long-term rehabilitation measures such as, reconstruction of houses in safer locations, provision of basic amenities like drinking water, sanitation, roads, education, viable livelihood systems, health care, and up gradation of farming technologies in the relocated areas.

Check Your Progress: 2

Note: 1) Use the space provided below for your Answers.
2) Compare your answers with those given at the end of the unit.

1) Explain the following terms:

- a. Rehabilitation
- b. Environmental migrant

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3.4 ENVIRONMENT, TECHNOLOGY AND SOCIETY

Need of development has led to industrialization, trade and urbanization, which are observed as symbols of national progress. Human population has also increased over time with associated technological advancement in medicine and related fields. The developmental needs are growing day by day because of the increasing demands of the society and man is continuously encroaching upon nature for satisfying his needs of water, food and energy.

Technology has helped humans ease out their work. Before understanding the relationship between Environment, technology and society, there is a need to focus on the basic definitions.

Definitions:

- **Technology**

Technology involves the systematic application of scientific knowledge, art, tools and materials for the extension of human facilities. Technology is the source of change in society.

- **Environment**

Environment is defined as the immediate surroundings of living organisms that include air, water, soil etc. that supports life and sustains various human activities. The surroundings comprises of

- Biotic/ living: Plants, animals, microorganisms
- Abiotic/ non-living: Physical phenomena such as land, water, air etc.
- **Society**

Society is group of people living together in communities in an organized manner, wherein the needs and development is sustained by the community. It is a permanent organization in which individuals are bonded together by social relationship.

There is a significant relationship between environment, technology and society, which can be demonstrated by Figure 3.6.

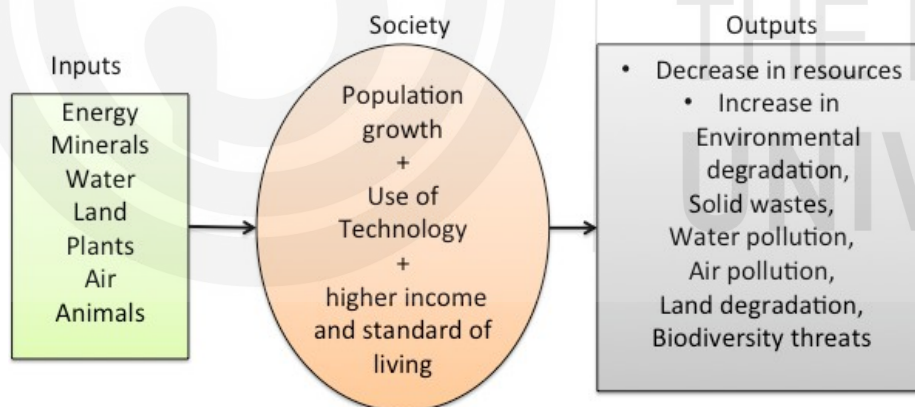


Figure 3.6: Connection of resources, society, environment and technology

- All societies have an ecological basis. Ecology is composed of physical as well as biological systems involving human beings.
- The term 'ecology' denotes the web of physical and biological systems in conjunction with the processes involved, of which humans are only one element.
- The ecology of a place is primarily affected by the interaction between its geographical and hydrological components. Besides these, technological interventions also affect the ecology of a society because of which environmental problems such as land degradation and pollution are increasing.

- Ecological changes are disturbances to the environment caused by human interference, natural disasters or animal interventions. It is often difficult to separate and distinguish between the natural and human factors that bring ecological change as both cause massive destruction.
- The interaction between environment and society is shaped by social organization/ social structure. This is discussed in the section on Relationship between environment and society.
- Environment acts as a service provider of valuable resources to the society without which it is impossible to address the needs of mankind. Economic activity is supported by rules and condition created by human societies as to how and by what process they plan to generate income. The relationship between environment, society and economy can be observed in Figure 3.7.

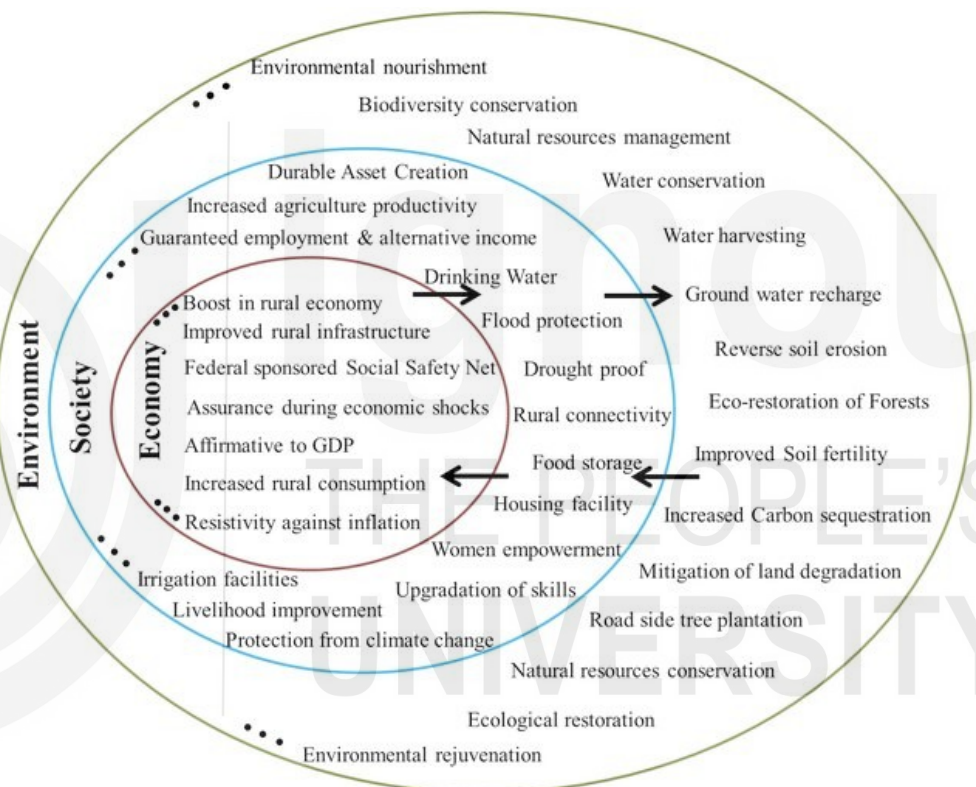


Figure 3.7: Interrelationship between Environment, Society and Economy (Adapted from Dandabathula et al, 2019)

It can be observed that there is a continuous exchange of resources between the Environment, society and economy as demonstrated through the arrows. For example, measures to protect environment may lead to improved soil fertility. This can further lead to increased food production and thus better food storage facilities. This shall further enhance consumption boosting the rural economy. The boost shall lead to employment and income creation thus generating more awareness regarding natural resources management issues.

Relationship between environment and society

- Human beings, society and Environment are mutually dependent on each other. The relationship is represented in Figure 10.8. It can be seen that societies are dependent on the environment for basic resources such as

food and fuel. In order to get these resources we need to work. So, indirectly environment provides us with economic opportunities.

- These economic opportunities are constrained by social structure (Pretty et al, 2007). Social structure is the stable arrangement of institutions where humans interact with society in order to stay together. One example of social structure is market behavior. These structures change with change in environment.
- In case of some natural hazards/ calamities, the cost of materials in market increases because of change in demand/supply ratio. This imbalance in the ratio might bring in competition between individuals changing their societal behavior. So, while human actions can impact environment and society, environmental calamities can also affect social structure leading to change in human behavior.
- Use of technological interventions such as usage of machines to ease out work can lead to technological hazards such as pollution, which can further harm the environment.

Culture can also have an impact on the society and in turn on the environment. For example, a culture that focuses on women education makes the society more modest, caring and brings in a value system associated with resources.

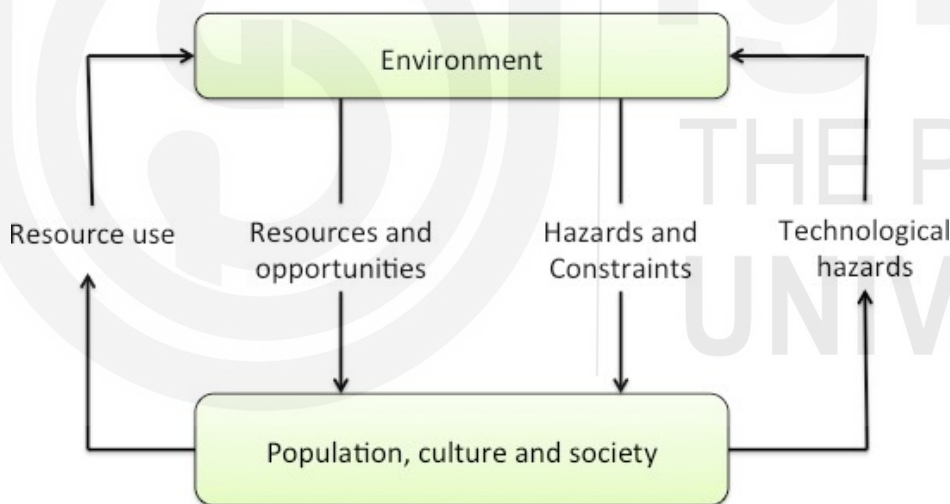


Figure 3.8: Relationship between environment and society

Risk society

- Risk society is a sociological concept introduced by Ulrich Beck. According to him, technological advancements can make our lives easier but may create a risk to the environment.
- Risk society may be considered as a society in which the future holds a risk due to some environmental/ industrial calamities.
- For instance, incidents like nuclear disaster of Chernobyl and industrial accidents like Bhopal depict the dangers inherent in industrial environments and demonstrate the associated risks that human population can be subjected to.

- We consider ourselves as living in risk societies because human relations with the environment have become increasingly complex in modern society due to spread of industrialization.
- The complex industrial technologies and modes of organization require sophisticated management systems, which are often fragile and vulnerable to error. Any error in handling or management could lead to severe consequences. Hence, it is difficult to predict the risk involved in any complex technology and careful handling is the only viable option.

Technology and Environment

- Technology may have positive as well as negative impacts on the environment. If we consider agricultural production for instance, it can impact the ecosystem in both positive and negative ways. Technologies used in agricultural production may have impacts on soil fertility, biodiversity, water or climate. Fertilizers may be added to improve soil fertility but may negatively impact the health of the farmer who comes in contact with it. These fertilizers may also be harmful for the animals living within the agricultural land ecosystem. In such cases, doing Cost-benefit analysis, bringing in more of traditional practices such as mixed cropping or crop rotation can be extremely helpful.
- Another example of impact of technology can be observed in our daily lives when we use gadgets such as smart phones, television or other electrical appliances. Life has become easier and advanced but we do not focus on the amount of electrical energy that is spent for functioning of these appliances. Electrical energy needs utilization of non-renewable resources such as fossil fuels, which are depleting day by day. Thus, it is important that solar or wind power may alternatively be utilized for electricity production. Understanding that development is necessary, but it should be achieved by minimizing the negative impact on environment is essential.
- Being aware of technological methods that can reduce environmental impacts can further solve the problem. For example, if a new product is to be designed, its life cycle should be assessed. The manufacturer can consider calculating the pollution it can cause at every stage starting from the raw materials, the energy needed for its production, industrial treatments, transportation technologies to take it to market and finally its disposal. By this assessment, one can try to reduce the environmental impacts by either utilizing eco-friendly materials or utilizing those industrial techniques that generate lesser pollution. Thus, sustainable practices should be followed and promoted wherever possible.

Check Your Progress: 3

Note: a) Write your answer in about 100 words.

b) Check your progress with possible answers given at the end of the unit.

1) Give some examples how technology affects the environment.

- 2) Explain how Environment and society are interrelated.

3.5 LET US SUM UP

In this unit, we have discussed that the pioneering report ‘The Limits to Growth’ published in 1972, suggested that the population growth and resource consumption needs to be significantly reduced. The report predicted that the resources are limited and if the current trends of industrialization stay as such, the resources might be exhausted by next 100 years. Though there were certain criticisms to the report as many scientists could not think that there could ever be any limits. But, the present patterns of resource depletion agree well with the report. If we consider facts from the history pertaining to environment, Marx could be considered as a revolutionary social thinker who could think well ahead of his times even regarding environment. He stated that people, especially from the labor class should stay connected with the nature as they are an important link between nature and levels of production. He emphasized that capitalism had created a rift between man and nature and there was a need to heal those wounds and work towards a sustainable future. In India, Mahatma Gandhi, a visionary leader, advocated for using only the right amount of resources that were required. He emphasized that “Greed could never be satisfied”. Gandhi supported the craftsmanship of tribal communities and requested people to adopt practices such as weaving on charkha as he felt that this would give impetus to the indigenous technologies. It is well known that Education is the major way to get wisdom and knowledge and another visionary Jyotirao Phule encouraged education of Dalits, minorities and women to liberate them from conventional shackles of society.

It has been observed that throughout history, especially through the phase of industrialization, environmental concerns have taken a backseat and caused major problems such as air pollution, global climate change, water pollution and generation of hazardous waste. This has further led to Resettlement and Rehabilitation of groups of people directly/ indirectly affected by these activities and is sometimes necessary as a remedial measure of such environmental problems. While relocating, these people should be provided with benefits that they were enjoying earlier to ensure an easy transition.

It is a well-known fact that society and environment are in constant exchange of resources or energy. It is the prerogative as well as duty of society to manage its resources wisely. Technology has benefited the society in terms of reducing labor. However, its extensive usage has created some unforeseen environmental problems. Thus, the key is to strike a balance between the two to ensure a healthy society and a favorable environment. This balance could be achieved by practicing principles of sustainability and revisiting the thoughts of social thinkers like Karl Marx or Gandhi so that the future is bright for the upcoming generations.

3.6 KEY WORDS

Sustainability: The ability of sustenance of future generation by wisely using the present resources. Resources include natural, social as well as economic resources.

Capitalism: It is an economic setup when the factors of production such as capital goods, labor and resources lay under the private ownership. This term is mostly used when majority of industrial sector is run by private sector, driven by the motive of profit with less regard to a larger social good or equitable distribution of resources as found in socialist economy.

Industrialization: It is the process in which a society or an economy undergoes transition from agriculture to manufacturing or production of goods. Machines that can lead to mass production replace manual labor. As a result, there is often better economic growth, better division of labor and better use of technology in certain sectors. However, it may lead to problems such as environmental pollution and heterogeneous distribution of power and wealth amongst a particular section.

Satyagraha: It was a tool to resist evil and injustice through non-violence. It was introduced by Mahatma Gandhi and was frequently practiced by him throughout Indian struggle for independence. It still hasn't lost its significance and is quite instrumental while raising voice against prejudices and discrimination.

Displacement: When group of people are forced to move within or across border due to conditions like political unrest or some environmental calamity is termed as displacement.

Relocation: State- organized people movement undertaken with informed consent, wherein, state takes charge of re-establishing living conditions of affected people. This is generally done over because of an upcoming state developmental projects and the shifting might/ might not be permanent in nature.

Pollution: It is a phenomenon in which contamination of the entities of environment (like land, water or air) occurs above a particular threshold level at which they become harmful to the health of human beings. Noise and light can also be considered as pollutants if they exceed the tolerance limits of humans or other living organisms.

Global Warming: When the content of green house gases such as CO₂, increases in atmosphere, they tend to trap in frared solar radiation that is exiting the earth surface. The gases act as a blanket and due to this heat trapping, the temperature of earth rises further. This process is known as global warming. Global warming is that extensive now, that the land and ocean surface temperature for January 2020 has been observed to be 1.14°C higher than the 20th century average (Source: NOAA-NCEI State of the Climate: Global Analysis).

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3.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check your Progress: 1

1. The Limits to Growth report consisted of five variables, which were industrialization, population, food, use of resources and pollution. Conclusions of Report were:
 - 1) Data was modeled till 1970 and based on the trend, the scenarios of variables were predicted for year 2100. It was concluded that if population increase and resource consumption occurs at the same pace, the economy would reach a stage of collapse.
 - 2) The only way to protect the ecological and economic entities was to switch to sustainable practices.
 - 3) Maintaining a balance between ecological and economic entities and reducing the pace of industrialization was the only way to succeed.
2. Some of Gandhian perspectives on environment were:
 - **Wise use of resources:** According to Gandhi, a man should keep limited number of material needs and have plain living. He said that comfort is necessary but not to an extent that it becomes greed. Plain living and high thinking of mankind was his ideology of living.
 - **Stress on Environmental problems:** The patterns of growth and development followed at that time because of industrialization was considered as a disease by Gandhi. He consistently focused on following indigenous practices to maintain the ecological balance.

Check your Progress: 2

- a) **Rehabilitation:** It is resettlement of a person that incorporates his/her overall well-being and takes care of his/ her economic stability, accommodation and culture.
- b) **Environmental migrant:** A person who voluntarily resides outside or within his/her native country for reasons of changes to the environment, sudden- or gradually-onset.

Check your Progress: 3

1. Industrialization along with technology has had a negative impact on environment. Some examples are:
 - Generation of green house gases such as CO₂, methane, nitrous oxide and ozone through activities such as combustion of fossil fuels has led to global warming which has resulted in major devastations of earth. Some of the devastations include melting of glaciers, sea level rise and change in rainfall patterns.
 - Air and noise pollution resulting from industries. When coal is burnt in thermal power plants, sulphur dioxide is released which tends to pollute the environment.
2. Society is dependent on the environment for resources. Society gathers its supplies from environment and in turn is responsible for its degradation in terms of pollution of land, air and water. Environmental calamities such as cyclones and earthquakes and phenomenon such as global warming due to rise in concentration of green house gases can also have huge impacts on societies. Global warming may further lead to submergence of low-lying lands due to rise in sea levels forcing people to migrate to different locations. Similarly, deforestation leads to disruption of ecosystems and affects animals and biodiversity dwelling in that area.

UNIT 4 DEMOCRACY, CIVIL SOCIETY AND ECOLOGY

Structure

- 4.0 Introduction
- 4.1 Learning Outcomes
- 4.2 Historical Development of Science and Technology
 - 4.2.1 Industrial Revolution in Indian Context
- 4.3 Technology Adaptation
 - 4.3.1 Third World Countries and Adaptation of Technologies
- 4.4 Let Us Sum Up
- 4.5 Key Words
- 4.6 Further Readings
- 4.7 Check Your Progress: Possible Answers

4.0 INTRODUCTION

Environmental governance is a grouping of ethics and morals that have been getting illumination of their sparkle from the civil society linked with ecological liability, governmental transparency, democratic institutions, the rule of law and environmental integrity to '*We, the People of India*'. The environmental governance of a system based on involvement, responsibility, and justice confirms the broad political and socioeconomic issues of marginalized segment of society. There is an increasing importance on governance as a vital aspect of environmental conservation which calls for active participation of civil society. There is an urgent need to tailor the discussion and knowledge around these themes with a focus on rationalizing the functional role of the civil society organizations in context of environmental governance. It mainly underlines the need for suitable government-civil society relations. The role of civil society is gradually becoming prominent in context of environmental governance. Therefore, there is a growing need for a thoughtful debate on this subject as the norms of environmental governance express a resolution to have a charter of human rights for human existence with pride while bringing the socioeconomic development in the orbit of environmental justice. There are impending challenges like the execution of environmental policies and integration of socio-environmental concerns into sustainable development framework while focusing on gender, human rights and rule of law in environmental governance. Hence, the role of civil society in firming the environmental governance, nurturing human rights, sensitizing people about the environment, and preserving ecosystems has become more serious than ever before. This chapter deals with historical development of science and technology and industrial revolution in Indian context.

4.1 LEARNING OUTCOMES

After studying this unit the learner will be able to:

- describe the historical development of science and technology;
- explain industrial revolution in Indian context; and
- describe the third world countries and adaptation of technologies.

4.2 HISTORICAL DEVELOPMENT OF SCIENCE AND TECHNOLOGY

The history of Indian science and technology can be traced from the primitive human activity during the Indus Valley Civilization to the early states and kingdoms. Being one of the eldest civilizations of the world, the Indian civilization has a powerful institution of science and technology. India is known to be the land of saints and sages who were known for their technical knowhow and scholastics. India has been actively contributing to the field of science and technology long before the modern laboratories were established.

The theories, procedures and principles evolved by ancient Indian saints, scientists, physicians and mathematicians have supported the fundamentals of contemporary science and technology. While some of the pioneering contributions have been acknowledged, some were faded into oblivion. The area-wise historical development of science and technology is presented in the following paragraphs:

Indian civilization has a powerful institution of science and technology. India is known to be the land of saints and sages who were known for their technical knowhow and scholastic approach.

1) Mathematics and Physical Sciences: Vedic people had special interest in mathematics and astronomy. The people of the Indus valley evolved a standardized method of using weights and measures, which was evident from the excavations made at various archeological sites. The significant contributions which changed the scenario of science and technology as a whole were:

- **Concept of Zero:** Notion of ‘Zero’, the mathematical digit, by the great Mathematician Aryabhata, was one of the most important developments.
- **Decimal System:** The method of expressing numbers by decimal system was given by India, which eased the calculation and made the use of arithmetic much faster and easier.

- **Numeral Notations:** A system of symbols for every number from one to nine was evolved as early as 500 BCE and was later adopted by Arabs and western world who named it as Arabic numerals. Numerous Vedic texts mention about the science of numbers and other sciences, for instance, the *Chandogya Upanisad* and *Mundaka Upanisad*.
- **Binary Numbers:** The concept of Binary numbers, which formed the basics of computer programs, was first described by Pingala, the Vedic scholar, in his book *Chandahsastra*.
- **Chakravala Algorithms:** A cyclic algorithm to solve indeterminate quadratic equations was developed by Brahmagupta, one of the famous mathematicians of the 7th century and later generalized and refined by Jayadev and Bhaskara II.
- **Knowledge of Measurements:** The early excavations have yielded linear measures of ivory and shell, which revealed that the knowledge of measurement was well-developed and precise.
- **Theory of Atom:** Kanad, one of the distinguished scientists of ancient India devised the atomic theory centuries before John Dalton. He speculated the existence of *anu* or a small indestructible particle, much like an atom.
- **Heliocentric Theory:** Mathematicians of ancient India applied mathematical knowledge to make precise astronomical predictions. The most significant among them was Aryabhatt whose book represented the zenith of astronomical knowledge. He proposed that the Earth is round, revolves on its own axis and spins around the Sun, i.e. the heliocentric theory. He also forecasted about solar and lunar eclipses, duration of the day and distance between Earth and Moon.
- **Kharosthi numerals:** These numerals are found to occur in Asokan, Saka, Parthian, and Kusana inscriptions from 4th century BC to 2nd century AD. Strokes and crosses were used for the first eight digits.
- **Arithmetic:** The method of calculation of cube root of any integral number was evolved by Aryabhatt.
- **Algebra:** The early stages of algebra, or more appropriately, the geometrical method of solving algebraic problems, can be traced to Sulvasutras of Apastamba, Baudhayana, Katyayana, Manava, and a few others. The differentiation of algebra as a distinct branch of mathematics took place from the time of Brahmagupta.
- **Geometry:** Like other branches of mathematics, geometry in India was developed in the course of dealing with practical problems.
- **Astronomy:** There is substantial material on astronomy in Vedic Samhitas. Much progress was made during the Brahmana period when astronomy was considered as a separate science called Naksatra-Vidya, the science of stars. There is mention of astronomical phenomenon in Vedic Literature. According to Rig-Ved, the universe comprises of *prthivi* (earth), *antariksa* (sky, the

region below the stars), and *dyaus* (heaven). The *Satapatha Brahmana* describes earth as *parimandal* or sphere. There is evidence in Rig-Ved about the axial rotation and annual revolution of the earth. According to Rig-Veda, sun is the maker of day and night, twilight, month, year and seasons.

2) **Metallurgical Science:** Several advances were made in the field of metallurgy, for instance:

- **Wootz Steel:** Wootz steel, a crucible steel was known to ancient Indians, mainly the Tamils. The steel was used to make the famed Damascus swords that could cut a free-falling silk scarf or a block of wood with the same ease.
- **Zinc Smelting:** India pioneered in Zinc smelting by using the distillation process, an innovative method of ancient alchemy. Zawar in Tiri Valley of Rajasthan is world's first known ancient zinc smelting site.
- **Seamless Metallic Globe:** One of most remarkable feats in metallurgy was the first seamless celestial globe of Kashmir made during the reign of Emperor Akbar. In a major accomplishment in metallurgy, Mughal metallurgists established the method of lost-wax casting to make other globe masterpieces.
- **Iron-Cased Rockets:** The first iron-cased rockets were developed by Tipu Sultan of Mysore in 1780s. These were effectively used against the British East India Company during the Anglo-Mysore Wars. He manufactured long iron tubes, filled them with gunpowder and fastened them to bamboo poles to create the precursor of present day rocket. With a range of about 2 km, these rockets were the finest in the world at that time and caused fear and confusion. Archaeological diggings have revealed that Harappan people were master craftsman and had knowledge of metallurgy. Even they made bronze by mixing copper and tin. India has ancient gold mines in Maski region of Karnataka. While the Indus Valley civilization belonged to the Bronze Age, its descendant, the Ganges civilization, which arose in the first millennium BC, belonged to the Iron Age. Right from the 17th century, several European travellers documented India's iron and steel-making furnaces. Most of India's metal was confined to the tribes, for instance, the Agarias of Uttar Pradesh and Madhya Pradesh were reputed ironsmiths. Many ancient texts, for example, Kautilya's Arthashastra refer about the metals. Metal currency was minted in India before 5th century BC, with coinage of silver and copper, bearing animal and plant symbols on them.

3) **Medical Sciences:** Even before the birth of Hippocrates, Charak authored an introductory text, *harka Samhita*, on Ayurved. Charak was the first physician to give the concept of digestion, metabolism and immunity in his book, which was later translated into many foreign languages. The scope of Ayurved is not limited to physical health alone. It seeks to stimulate physical, mental, and spiritual health in context of

man's interaction with his environment. The scope of Ayurved, in general, covers:

- a) Cosmological suppositions about the basic link between matter and life;
- b) Biological theories concerning:
 - embryonic conception,
 - body, life, and soul, and
 - rules of genetics;
- c) Physiological and pathological concepts;
- d) Food and rules of health and long life;
- e) Diseases, their diagnosis and treatment;
- f) Poisons and antidotes; and
- g) Ethics

The humeral theory of Ayurved propounds that *Vayu* (wind), *Pitt* (bile), and *Kapha* (phlegm) are the three basic elements stimulating, supporting, nurturing, and upholding the life-principle. Ayurved deals with:

- Kaya cikitsa (therapeutics)
- Salya tantra (major surgery)
- Salakya tantra (minor surgery)
- Bhuta vidya (demonology)
- Kaumarabhrtya tantra (pediatrics)
- Agada tantra (toxicology)
- Rasayana tantra (geriatrics)
- Vajikarana tantra (virilification)

Rasacikitsa was later incorporated in Ayurved in 1300 A.D. Numerous preparations of mercury, iron, copper, and other metals as formulated in alchemy were used in Ayurvedic treatment. Written by Sushrut in 6th Century BC, *Sushrut Samhita* is one of the most comprehensive textbooks on ancient surgery. The text mentions various illnesses, plants, preparations, cures and complex techniques of plastic surgery. The *Sushrut Samhita's* most famous contribution to plastic surgery is the reconstruction of nose, known also as rhinoplasty. The first cataract surgery is said to have been performed by ancient Indian physician Sushruta, way back in 6th century BC. To remove the cataract from the eyes, he used a curved needle to release the lens and push the cataract out of the field of vision. Sushrut's surgical works were translated to Arabic language and later introduced to the Western world.

- 4) **Engineering and Architecture:** The accomplishments in the field of engineering started in the proto-historic times, from the 3rd millennium B.C. during Harappa period. A number of towns and cities, called *janapadas*, of great significance were developed before the 7th century

BC. Noteworthy among them were Ayodhya, Varanasi, Kampilya, Mathura, Mithila, Sravasthi, Ujjayini, and Vaisali. There is mention of dams and bridge construction in Kautilya's *Arthashastra*. The construction of suspension bridges using interwoven bamboo and iron chain was evident during the 4th century. The Buddhists presented their engineering skill in the construction of religious edifices like *Stupas*. The spread of Buddhism motivated the setting up of monasteries. The design of a rock-cut *vihara* was the first planned work by an architect or master craftsman. Gupta period (A.D.300-600) viewed the early stages of systematic temple construction. The rock-cut temples, excavated by Asoka in Barabar and Nagarjuni hills, illustrate the basic procedures of rock-cut architecture. The Kailasa temple at Ellora stands matchless as a monument to the creativity and expertise of Indian rock-cut architecture.

- 5) **Agriculture:** Farming came into being when the early man gave up the nomadic lifestyle and started a settled life in the favorable climate and topography. Although it has not been well-established when the natives of India opted for farming as the primary occupation; but, the origin of agriculture can be traced back to the Indus valley civilization. Thus, at least for the last 4,500 years, the Indian society has mainly been agrarian. There is archeological evidence of animal-drawn plough which dates back to 2500 BCE in the Indus Valley Civilization.

Innovative irrigation and water storage systems including artificial reservoirs were developed by during the Indus Valley Civilization, for instance, Girnar dated to 3000 BC, and early canal irrigation system from 2600 BC. During the medieval period, the agriculture pattern was more or less the same as that of primitive India; however, some significant changes occurred with the introduction of new crops, trees and horticultural plants by the foreign traders. The main crops were wheat, rice, barley, millets, pulses, oilseeds, cotton, sugarcane and indigo. The Western Ghats continued to yield black pepper of good quality and Kashmir maintained its tradition for saffron and fruits. Ginger and cinnamon from Tamil Nadu, cardamom, sandalwood and coconut from Kerala, gradually became popular. During the sixteenth and seventeenth centuries, tobacco, chilies, potato, guava, custard apple, cashew and pineapple were introduced in Indian Sub-continent. Later, the production of opium started in Malwa and Bihar regions. Likewise, there were new innovations in agricultural practices, which were adopted successfully, for instance mango-grafting was introduced in Goa in the mid-sixteenth century. Wells, tanks, canals, *rahat*, *charas* and *dhenkli charas* were used for irrigation. Cultivation of fruit trees was promoted in the imperial Mughal gardens.

- 6) **Biodiversity and Environment:** Ancient Indian insight of 'life' revolves around all biotic and non-biotic components of environment. According to ancient Indian thought, the nature is composed of five basic elements, i.e., earth (*shristi*), water (*apah*), fire (*teja*), air (*vayu*) and space (*vyoma*). The five elements are all the facets of divinity. Since the five elements are of divine origin, our forefathers have adored them as the personifications of divinity. They worshipped them as goddess earth,

many goddesses signifying rivers, the fire god, the wind god, and the god of ether. Primarily, our ancestors venerated earth as *Bhu Mata* (Mother Earth). Since the food obtained from its lap helped in continuing the life, they worshipped it first. The nature or *prakrti*, as known in Indian context is believed to be the supreme mother of entire creation. Ever since the advent of life on the planet earth, water is the source of life and is most indispensable for survival. Even the life is believed to have originated in water itself. Presence of life on earth was responsible for origin of life. In *Rig Ved*, water is placed in the category of nectar. In Vedic literature, the term '*Arankaya*' is used for forest, which signifies hermit of peace. *Vriksh Ayurved*, the science of plant life, scripted in ancient India as a special branch of knowledge, primarily describes the useful features of plant science. Forests of different types have been described as:

- *Shrivan*, the forest accountable for prosperity;
- *Tapovan*, the forest meant for contemplation by seers and sages; and
- *Mahavana*, the big forest where all species find shelter

Plants and animals find mention in Vedic literature. Plants are categorized according to their morphological characters, medicinal and food value.

The sequence of *Dashavatara*, ten incarnations of Lord Vishnu explains Darwin's theory of evolution. Like the evolutionary progression, the first incarnation of God is *Matsya*, a fish, followed by *Kurm*, the turtle, later *Varaha*, the boar, *Narasimh*, a man-lion being, *Vaman*, the dwarf and rest four the human forms. While comparing the biological events at different geological time scale, it is seen that the life first originated in ocean in the form of fish during the Cambrian period. In geology, the reptiles come at second place in evolution which is believed to be originated in Mississippian period just after amphibians. In tertiary period, all mammals originated. The incarnation of *Vaman* may be related with the first human being who originated during the Pliocene period. *Parsuram* has similarities with the first modern man who originated during the quaternary period or the man of Iron Age. While comparing Darwin theory of evolution with ten incarnations, similarities can be seen. Some contemporary interpreters talk about Lord Vishnu's ten incarnations in an ultimate order, from simple life-form to more complex, and view these incarnations in light of modern theory of evolution. Such explanation was first proposed by Theosophist Helena Blavatsky.

After deriving motivation from the mythology, rulers selected animals and plants as the insignias of their dynasties and used them in art and architecture. Indian mythology has deep influence of Hindu-Buddhist theory which asserts that soul passes through different stages or modes of existence, i.e. animals, human and superhuman. The place of plants & trees in ecology has been well understood by the Indians, which is evident from mythological literature.

Later during the medieval period, Science and Technology in India developed two aspects: one based on the earlier traditions and other with new influences which were consequent of Islamic and European impact. With the

passage of time, the primitive Sanskrit manuscripts were translated during the Muslim period. The intellectuals like Govinda, Sankaranarayan, Aryabhata II, Sripati, and Satyanand flourished and created an intellectual atmosphere for the growth of astronomy in India. South India played a significant role in refining astronomy. Mughal metallurgists initiated the method of wax casting. The instruments and observational systems used at Mughal observatories were mostly derived from the Islamic tradition, with computational techniques derived from Hindu tradition. Chemistry during the medieval India was closely linked with alchemy which was an important part of Tantric cult. The alchemical philosophies and discourses of India found their way to China and Tibet. There is sufficient evidence about application of chemical knowledge and processes during the medieval period, mainly in the fields of metallurgy and metal work. During the medieval age, Indian medicinal science which was dominated earlier by Ayurved, observed changes. New medical practices, such as Unani system of medicine, reached India with Arabian, which brought changes in medical systems and pharmacology. During the medieval period, the study on fauna also made progress. The Mughals showed interest in animals, their habits, modes of life and their ecological distribution in and around the country.

During the Mughal Era substantial work was done in the field of mathematics, medicine, astrology, astronomy, and translations of various texts. Science was not patronized and chased thoroughly as a state policy by the kings as it was developed in Europe. Scientific institutions were not set up, nor were the students encouraged to go out for higher studies. The money was spent on constructing monumental structures, rather than creating scientific institutions. There was a significant change in the education system. The Arabic system was introduced and *Madrasas* were opened all over. The Mughals tried to introduce reforms. Later, several works were written in the fields of mathematics, chemistry, biology, astronomy and medicine. Most of the scientific work in this period was commentaries or explanations of earlier manuscripts. Several scientific works of astronomy, medicine and other sciences were translated from Sanskrit to Persian/Arabic and vice-versa.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) What is the contribution of mathematics and physical sciences to the modern world?

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4.2.1 Industrial Revolution in Indian Context

Industrial revolution is the process of transformation from a farming and handicraft-based economy to industrial production. Industrial revolution started in Britain in 18th century and later spread to rest of the world. The term *Industrial Revolution* was first proposed by the English economic

historian Arnold Toynbee to define Britain's economic development from 1760 to 1840. The main characteristics of Industrial Revolution were technological, socioeconomic, and ethnic. The technological changes included:

- 1) Use of new materials, mainly iron and steel;
- 2) Use of new energy sources, including coal, electricity, petroleum, etc;
- 3) Discovery of innovative machineries, for instance power loom, which increased the production with a smaller expenditure of human energy;
- 4) Introduction of factory system with division of labor;
- 5) Development in transportation and communication, including the steam locomotive, automobile, airplane, telegraph, radio, etc.; and
- 6) Increased application of science in industry

The technical revolutions made possible the increased use of natural resources and production of manufactured goods. Besides this, there were new developments in non-industrial domains, viz.:

- **Industrial Revolution is the process of transformation from a farming and handicraft-based economy to industrial production.**
- **The period of First Industrial Revolution (1760 to 1830) was largely confined to Britain.**
- **Two Englishmen, William and John Cockerill, brought the Industrial Revolution to Belgium**

- 1) Improvement in agriculture;
- 2) Economic transformations that caused broader distribution of wealth;
- 3) Political changes revealing the shift in economic power, besides new state policies matching the requirements of an industrialized society;
- 4) Comprehensive social change, comprising of growth of cities, development of working-class movements, and rise of new arrangements of authority;
- 5) Cultural change, where workers acquired new and unique skills and their relation to occupations transformed; rather than being craftsmen working with hand tools, they became machine operators; and
- 6) Psychological change, which improved the self-confidence to use the resources.

The period of First Industrial Revolution (1760 to 1830) was largely confined to Britain. Aware of their head start, the British banned the export of machinery, skilled workers, and manufacturing methods. The British

domination could not last forever, as some people saw lucrative industrial breaks out of country, while mainland European businessmen sought to attract British know-how to their countries. Two Englishmen, William and John Cockerill, brought the Industrial Revolution to Belgium and it became the first country in Europe to be economically reformed. Like its British predecessor, the Belgian industrial revolution focused on iron, coal, and textiles. France was more slowly and less systematically industrialized than either Britain or Belgium. France was engrossed in its revolution, and the ambiguous political situation disheartened large investments in industrial revolution. By 1848, France became an industrial power, but, despite excessive growth under the Second Empire, it remained behind Britain.

Other European countries lagged far behind. Political situations in other countries delayed the industrial expansion. The rise of US industrial power in 19th and 20th centuries outshined European efforts and Japan joined the industrial revolution with outstanding success. Mid-20th century viewed the expansion of industrial revolution into previously non-industrialized areas such as China and India.

Despite substantial overlying with the “old,” there was increasing indication for a “new” Industrial Revolution in the late 19th and 20th centuries. In terms of basic materials, modern industry started to exploit natural and synthetic resources not utilized yet. Together with these, there were improvements in machines, tools, and computers which gave rise to automatic factory.

Industrial revolution, a period from 1770 to 1850 is described by a change in the manufacturing process through the use of machinery and other inventions. Great Britain was the hub of industrial revolution. Britain's East India Company collected revenue and raw material from affluent areas of East India and sent money and material to Britain. Moreover, after goods were manufactured in Britain, these were brought to India for sale. Thus, India served to expand Britain's sale market for manufactured goods. India, however, was not benefited from industrial revolution in the manner the Great Britain was. Indeed, the Indian economy, which was once prosperous, was shattered by industrial revolution. Before industrial revolution, India was world's leader in manufacturing and exporting textile. All through the industrial revolution, Great Britain subjugated India till India's economy collapsed. Although India had numerous economic activities that added to its economy, the textile industry was a leading contributor. Prior to industrial revolution, India had globally acclaimed cotton textile industry and shipped the same to different parts of the world. India's textile industry flourished because it had access to cheap labor. From 1750 to 1947, India experienced modernization of its economy in various areas including agriculture, factory production, finance, and even film production. Industrialization based on the modernization of India's economy can be understood in four phases:

- Primary phase, linked with the extraction and production of raw materials, and involves farming and mining;
- Secondary phase, which takes the materials obtained from the primary sector and converts raw materials into final goods for example textiles and automobiles;

- Tertiary phase includes the provision of services such as banking, finance, and cinema to consumers and businesses; and
- Quaternary phase, which appears recently in the modernization process.

I) Development of Primary Sector

Agriculture: As compared to other countries, India had more arable land reserved for main food crops, including wheat, barley, maize, millet, sorghum, and rice. The increase in available farmland was mainly due to the development of irrigation and canal networks. During the period between the two world wars, i.e. 1918 to 1939, agriculture declined. During the post- World War I, India started to produce more cotton than before. British exploitation of Indian raw materials, as cotton increased radically during the 1930s when the British economy suffered from great depression. At the beginning of 19th century, India became the supplier of cotton, for England's Lancaster cotton textile industry. India's cheap cotton and other raw materials were exported to Britain to help produce final goods in British factories.

Livestock, Meat, and By-Products: There was no significant change in the number of livestock in India from the late 1800s to 1947. From 1895 to 1947, the number started increasing. There were no systematic records of meat and byproduct outputs in India till the 1950s, but dairy products were produced in large quantities.

Mining: Coal mining was a profitable business during the colonial times. However, the British executed the Arms Act in 1878 to outlaw India's ownership of firearms and limited Indians from mining and several mines were closed during the British rule. Large-scale commercial coal mining in India started in 1774 under the East India Company in the Western bank of Damodar River. The introduction of steam locomotives in 1853, made possible the effective transportation of coal from the mines to urban centers and ports. This increased India's output of coal from 2,203 thousand metric tons in 1890 to 30,695 in 1947. Coal mining thrived during and after World War I. Likewise, India's output of iron ore increased significantly during the British rule.

II) Development of Secondary Sector

Textile Industry: India had to struggle against cheap British textiles before independence. Prior to British rule in the eighteenth century, Indians dominated the world textile trade. However, along with industrial revolution, the advent of spindles, looms, and new spinning processes made better textile producers out of the British. In India, the textile industry evolved from being a mere domestic industry to a top national industry far before the industrial revolution. Meanwhile, after the opening of looms and spinning mills, Lancashire was the center of cotton and fabric industry in Great Britain, out-competing India though lower production costs, larger supplies, and forced prices. Thus, the traditional textile industry of India underwent de-industrialization during the British rule. However, modernization of India's textile industry took place during the early 19th century; the first textile mill in the country

was established in Calcutta in 1818. A few years later, the first cotton textile mill of Bombay was established in 1854. In 1861, the first cotton mill was established in Gujarat region. By the end of the 19th century, there were 178 cotton textile mills in India. Today textile industry is one of the biggest industries of Indian economy with huge production capacity, and large pool of skilled and low-wage workers.

Automobile Industry: India's automobile industry did not flourish until after independence, the foundations of domestic carmakers such as Tata Motors and Hindustan Motors were set up prior to 1947.

Iron and Steel Industry: In 1870, James Erskine founded the Bengal Iron Works, the first step towards an iron/steel smelting industry ever taken in India. The two historic iron and steel industries of colonial period that still exist are Tata Iron and Steel Company, Ltd (TISCO) and The Indian Iron and Steel Company, Ltd. (IISCO)

III) Development and Tertiary Sector

Cinema: India is famous for its cinema industry, known as '*Bollywood*'. The first motion pictures received by Indians were soundless short films, which were shown at the Watson Hotel, Bombay in 1896. Hiralal Sen and Thanawalla were the two pioneers of Indian cinema, producing short films in Calcutta and Bombay in 1900. But, the turning point came in 1913, when Dada Phalke produced India's first silent film *Raja Harish Chandra*. During the 1920s, Indian cinema became a regular industry and came under business laws and regulations. New film makers gained fame during 1920s and the Indian sound films replaced the silent films in 1930s.

Banking and Finance: Modern banking systems and institutions developed in India in the late 18th Century. Allahabad Bank established in 1865, was the first fully Indian owned bank. This initiative was one of many to finance the profitable Indian cotton trade. During the early 20th century, the banking market expanded to include Punjab National Bank in Lahore and Bank of India in Mumbai. The *Swadeshi* movement was effective and influential during the early 20th century, and local Indian businessmen and political figures established indigenous banks for Indian community.

IV) Development of Quaternary Sector

Prior to British rule, there education system was good and students from all classes of society attended schools. Pre-British schools did not have to pay land taxes, but after the East India Company took over, Indian education system suffered a lot. During the 19th century the higher education and western universities emerged. These universities were however, truly the part of British struggles for efficient colonization. Although the British established many higher education institutes, the Indian public was ignored and literacy among the Indians was very low during British colonization. Thus, the quaternary phase was unsuccessful under British administration, whose educational policies were short-term, biased, restricted, and accessible only to the elite class. Before the industrial revolution, India was known for its cotton textile

industry; but, the things changed after the industrial revolution. Some of the impacts of industrial revolution were:

- Discovery of steam power from the year 1815 threatened the Indian textile industry.
- The inventions of efficient and effective power loom abridged the cost of production and increased the competition.
- Indian farmers were forced to cultivate cotton and cash crops in place of food crops so that it can provide raw material for industries as India was under British rule. This resulted in deadly famines in India.
- Industrial revolution had severe consequences on society. Due to industrial revolution, cities started getting overcrowded gradually.
- For increasing production, the factory owners used to hire unskilled labor, and hence in tender age many children were forced to work in the factories.

There were some positive effects of Industrial Revolution, for example, the introduction of assembly line and factories, electricity development, and railroads which contributed to faster and more efficient production of goods and materials. There were charitable funding of schools and the enactment of child labor laws. Thus, the industrial revolution improved both economic production and the introduced new lifestyle among common people.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.
2) Compare your answers with those given at the end of the unit.

1) What are the technological impacts of Industrial revolution?

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4.3 TECHNOLOGY ADAPTATION

Technology Adoption is a term that refers to the support, inclusion and usage of new technology in society. The process involves several stages, usually characterized by the group of people who use the technology. It provides prospects for governments to advance and modernize the delivery of public services. Certainly, the technological advancement is linked with globalization. The technological advancement has increased the rate of globalization, as the rapid development of internet produced an enduring networked globe. The technology adoption by the developing nations has had deep impacts on their economies, for instance, it has reduced national costs of production, created quality standards, and permitted people to communicate

from a distance. The spread of technology powered by the internet has resulted in positive cultural changes in developing nations. Faster communication has helped in improving the democracy and poverty alleviation. In order to compete in a high-tech marketplace, the developing countries need individuals with technical expertise. Technological innovations and new technologies are having major impact on production in the industrialized countries. However, the inadequate transfer of such technologies to developing nations can increase the technology gap and reduce their competitive capability. Hence, far-reaching policy changes and emphasis on technological development must be evolved by the developing nations to benefit themselves of the increasing prospects of innovative technologies.

4.3.1 Third World Countries and Adaptation of Technologies

The term Third World was originally coined at the time of Cold War to differentiate those nations that were neither associated with the West (NATO) nor with the East. Presently, the term is used to describe the developing countries of Africa, Asia, Latin America and Oceania.

Technology has shown an important role in the development of the Western World, though the Third World nations are still waiting to incorporate advanced technologies into their system. With globalization and cooperation from the developed nations, advanced technologies can be permeated into the Third World countries, which can help in their economic growth and development. By adopting new technology, Third World nations will have the chance to have a better link with rest of the world. During the last two decades, there has been a major technological innovation in the field of:

The term Third World is used to describe the developing countries of Africa, Asia, Latin America and Oceania

- Electronics and informatics (i-pads, mobile phones, laptops, wireless ear buds, micro-blogging, apps, smart TVs, android phones, drones, etc.)
- Biotechnology
- Genetic engineering
- Lasers and remote sensing
- Virtual payment systems
- Virtual shopping
- Virtual home assistance
- Self-driving cars

Case Studies

- I) **Breakthrough in Biotechnology:** A critical field in new technologies which is having a considerable, positive impact on developing countries

is biotechnology. This involves the manipulation of micro-organisms to produce a range of products, including foods, chemicals, medicines and industrial enzymes. Biotech applications have developed widely in numerous fields, particularly in genetics, with groundbreaking developments in DNA structure and recombinant DNA technology. These developments have opened up enormous avenues for large-scale manufacturing of genetically engineered products and materials.

Development in biotechnology is certainly having a significant impact in the developing countries and biotechnology is contributing to agriculture, pharmaceuticals, healthcare, industry and mining, etc. In agriculture, major enhancements in plants and crops can be realized through genetic engineering, including the implantation of nitrogen-fixation genes. Tissue culture and related techniques have brought about significant improvements in plant breeding and production capability. Modified species of plants have been developed which could survive in adverse climatic conditions. Livestock breeding has improved significantly through genetic applications. The production of proteins, conversion of biomass to food products and increase in food production has been achieved through biotechnology. Biotechnology applications in the developing nations would depend on the extent to which the local resources and aptitude directed for the purpose. There is scope and potential for each nation to undertake biotech research and tissue culture with regard to its own production urgencies and to link local research and production competence with those accomplished in other nations. For this purpose, it is important to create and sustain an environment that fosters indigenous creativity and innovation.

- II) Information Technology:** The use of Information Technology (IT) in developing nations has caught the attention of many in the recent years. However, the research in this area is largely uneven, and there is a need to combine efforts to make research more effective. While technology continues to thrive in the Western World, Third World and underdeveloped countries continue to struggle with the idea of integrating such technology into their culture. After radio and television, which took decades to permeate across the globe, the internet has boomed into the society within few years. Many developing nations don't welcome the idea of internet in their nations and have a negative view toward the impact of media on their society. Many nations don't have the resources to acquire such technology. With cooperation and regulation, the Third World and developing countries can use IT to their advantage. Developed countries will only benefit from global interaction. Bringing IT into Third World countries is not only useful for those countries; but, it is helpful for entire world. With increase in communication, these countries are able to advance their knowledge of medicine and other knowledge to help their cultures and communities survive. Brining IT into the Third World countries is helping them strengthen their economic status.

These have increasingly been applied to a wide range of economic activities. With the ongoing technological revolution, new needs and

demands are emerging and the structure of production and services is undergoing rapid transformation in the industrialized economies. Modern technologies are both knowledge and research-intensive and have inter-sectoral application. There is a close relationship and interface between these technologies, mainly because of the links through microelectronics, and the impacts of these technologies are probably very extensive in the field of production and services in the next few years. Though these expansions have taken place mainly in the industrialized economies, their universal nature will have a substantial consequence on all economies, including those of developing nations.

- III) Mobile Phones:** Dealers are continuously searching for the innovative ways to send the marketing messages. Their attention changed towards mobile phones as an innovative medium for marketing, owing to their mass usage. Mobile phone reached billions of people around the world. This facilitated use of short messaging, social networking, mobile advertising, and media sharing as advertisement podiums for reaching the consumers. Mobile phone advertising has turned out to be better marketing platforms. Thus, the Mobile phone not only provided opportunities for marketing but also the means of communication.
- IV) Skype:** Skype technologies offer VoIP services, which has revolutionized the VoIP market by offering free internet telephony to the customers. Skype offered its services based on the P2P technology which removed the brokers and the profits thereof were confined to the customers. Since its launch, Skype's software had been used by the people for various purposes.
- V) E-Choupal:** It is an innovative initiative of ITC Limited, an Indian corporation, to improve its marketing channel in agriculture. The business model was planned to accommodate farmers, mediators in the traditional model through IT. The main objective of e-Choupal is dissemination and disintermediation of price information. E-Choupal deals with various agriculture products, marine products and extended the basket to horticulture and spices. E-Choupal operates in Indian states. Later, ITC launched Choupal Sagar, a multipurpose retail outlet in the rural areas to open rural malls.

Thus, technology in the 21st Century has turned out to be vital for every individual and organization in the Western World; however, the Third World countries are looking for including the technology in their policy and culture. Indeed, the incapacity of evolving a policy to incorporate technology in the system is one of the main reasons for the weak economy of those nations. Technology transfer has taken place mainly among the enterprises in industrialized countries, and has mainly been limited to computers, semiconductors and other electronics and telecommunications. The transfer of such technologies to developing countries has just started because of small markets for electronic products and their limited usage in the developing nations. In the field biotechnology, technology transfer has taken place mainly regarding the high-yielding varieties of crops. Biotechnology developments, which were formerly in the public domain, are gradually been

privatized, through registration of patents and industrial property rights on new developments. Biotech developments are of great value and interest for the developing countries, mainly in agriculture and allied sectors. So far, there has been little biotechnology transfer, though some multinational corporations are getting involved in the same in many developing nations. In the area of new materials, there has hardly been any transfer of technology to the Third World nations, except for the production of chemicals. Largely, the transfer of new technologies to the developing nations has been very limited. It is only in some South East Asian economies that offshore production of electronics products has been initiated.

A classification can be done with regard to applications and usage of new technologies in the developing countries. The first group comprises of countries of South East Asia, which have fairly adjusted to the technological developments and have achieved substantial export capability in electronics goods, like desktop computers, digital exchanges, etc. The second group comprises some of the larger developing countries, including Brazil, China, India and Mexico, in which there has also been considerable growth of indigenous electronics production, and research in biotechnology. Among these countries, the development of informatics and biotechnology is taking place gradually and successfully. The third group comprises of a large number of developing countries, including most other Latin American, Asian and North African countries, in which there has been limited application or development of new technologies in terms of use or production and research. Lastly, the least developed countries, which have not experienced much impact of new technologies, except where offshore production facilities for electronics.

Presently, technology with its progressive characteristics is used as the primary source to keep people linked across the globe. By integrating the World Wide Web and permitting the communication, the developing nations will improve their economy and raise the standard of living. It is important that the people living in the Third World countries understand that incorporating technologies into their system will be beneficial for them. Technology adoption will help in:

- 1) **Economic Growth:** The technology will support growth theory in economics and will be one of the greatest solutions to improve the current economic situations of Third World countries.
- 2) **Employment generation:** In many Third World countries the unemployment rate is very high. Technology adoption can decrease the unemployment rate by creating more job opportunities in these countries. Moreover, by integrating technology in the Third World countries more IT companies will be opened and this will increase the job opportunities. With the help of technological transformation, the benefits will be amazing and it will create local technical skills in the Third World nations. Today's technology and its modern features can be utilized by each industry. The rise of new services is one of the benefits the advanced technologies offers to different industries. Nowadays, mobile phones and other computer software make those services accessible

online for everyone. Third World countries can benefit from these services through advanced technologies in diverse fields such as farming, healthcare, and education.

- 3) **Globalization:** One of the main aims to adopt technology by the Third World nations is that the present technologies have already been developed by other countries and the Third World countries have to acquire, integrate and operate them. Consequently, with technological adoption, the Third World countries will be able to develop and share the same conditions as the Western World. However, different techniques must be determined for this phenomenon to happen. One of the possible ways to infuse technology to Third World nations is globalization, which can be extended by advanced technological advent. Globalization can be achieved through communication, technical and scientific workshops that will provide the Third World countries a chance to adopt the current advanced technologies.
- 4) **Infrastructure Development:** One of the most common challenges that Third World Countries face while incorporating technology, is the lack of infrastructure that might affect the technology adoption. Thus, before instilling technology in the Third World Nations a sustainable infrastructure plan must be evolved that could support the advanced technologies. With planning and developing infrastructure, technology will help in achieving the sustainable economic growth and development in diverse fields, for instance education, industrial units, and roads.
- 5) **Better Education:** While integrating technology in the Third World Nations for economic development and growth, it will improve the education. In fact, education must be a priority that the Third World Nations require today to go to the next level and even eventually in achieving the status of developed countries. Better education can benefit the nations in improving healthcare services, political **stability**, and economic growth that will ultimately alleviate poverty. By integrating advanced technologies education can have superior outcomes. A number of students in the developing nations don't attend school due to lack of resources. Luckily, with the advent of technology in the education system those resources can no longer be an issue. While integrating technology in the education system, particularly in the Third World countries, it is essential to create a curriculum that will satisfy the needs of the students.
- 6) **Political Challenges:** As the adoption of technology seems to be viable, various challenges may be encountered that might avert this strategy. One of the possible challenges is that the people living in Third World Nations do not accept the influence of the Western World. Many of them resist the plan of technology adoption. Indeed, they find that technology adoption will have an impact on their culture and society. This can be handled by identifying suitable ways to help those nations realize the global benefits of integrating technology. While integrating technology in nations where the cultural heritage is important, ethical factors must be considered.

In brief, to infuse technology into the Third World nations, it is important to take socioeconomic and political factors into consideration. Technology adoption by the Third World nations can make a positive difference. With globalization and help of developed nations, the Third World countries can achieve economic development and growth. This suggests that despite various upcoming avenues and benefits, with the exception of some nations in South East Asia and large developing nations, the impact of technological developments and its adaptation is still in the budding stage in the Third World nations.

Check Your Progress: 3

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1) Enlist various technological developments in the last two decades.

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4.4 LET US SUM UP

There is an increasing importance on governance as an important aspect of environmental conservation which calls for active participation of civil society. The history of Indian science and technology can be traced from the primitive human activity during the Indus Valley Civilization to the early states and empires. The theories, procedures and principles evolved ancient Indian saints, scientists, physicians and mathematicians have supported the fundamentals of contemporary science and technology. The historical development of science and technology is in the fields of:

- Mathematics and Physical Sciences
- Metallurgical Science
- Medical Sciences
- Engineering and Architecture
- Agriculture
- Biodiversity and Environment

Later, during the medieval period, Science and Technology in India developed two aspects: one based on the earlier traditions and other with the new influences which were consequent of Islamic and European impact. During the medieval age Indian medicinal science dominated earlier by Ayurved, observed changes. Afterwards, several works were written in the fields of Mathematics, Chemistry, Biology, Astronomy and Medicine. Several scientific works of astronomy, medicine and other sciences were translated from Sanskrit to Persian/Arabic and vice-versa.

Industrial Revolution is the process of transformation from a farming and handicraft-based economy to industrial production. Industrial Revolution started in Britain in 18th century and later spread to rest of the world. The technical revolutions made possible an increased use of natural resources and mass production of manufactured goods. The period of First Industrial Revolution (1760 to 1830) was largely confined to Britain. Britain's East India Company collected revenue and raw material from the affluent areas of India and sent the money and material to Britain and after the goods were produced, these goods were brought to India for sale. India was not benefited from the Industrial Revolution in the manner the Great Britain was. Indeed, Indian economy, which was once affluent, was shattered by the Industrial Revolution. Industrialization based on the modernization of India's economy can be understood in four phases, viz. primary, secondary, tertiary, and quaternary.

The technology adoption by developing nations has had deep impacts on their economies, for instance reduced national costs of production, creating quality standards, and permitting people to communicate from a distance. Faster communication has helped in improving the democracy and poverty alleviation. However, the inadequate transfer of such technologies to developing nations can increase the technology gap and reduce their competitive capability. By adopting new technology, Third World nations will have the chance to have a better link with rest of the world. Technology has shown an important role in the development of the Western World, but the Third World nations are still waiting to incorporate advanced technologies into their system. During the last two decades, there has been a major technological innovation in the field of electronics and informatics, biotechnology, genetic engineering, lasers and remote sensing, virtual payment systems, virtual shopping, virtual home assistance and self-driving cars. Technology in the 21st Century has been a vital for every individual and organization in the Western World. Technology transfer has taken place primarily among enterprises in industrialized countries, and has mainly been limited to computers, semiconductors and other electronics and telecommunications. It is important that the people living in Third World countries understand that incorporating technologies into their system will be beneficial for them. This will help in:

- Economic Growth
- Employment generation
- Globalization
- Infrastructure Development
- Better Education
- Political Challenges

In short, to infuse technology into Third World nations, it is important to take socioeconomic and political factors into consideration.

4.5 KEY WORDS

Industrial Revolution: It is the process of transformation from a farming and handicraft-based economy to industrial production. Industrial Revolution started in Britain in 18th century and later spread to rest of the world.

Third World Countries: The term is used to describe the developing countries of Africa, Asia, Latin America and Oceania.

Technology Adoption: Technology adoption refers to the appreciation, inclusion and usage of new technology in society. The process involves several stages, usually characterized by the group of people who use the technology. It offers prospects for governments to advance and modernize delivery of public services.

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4.7 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check your progress: 1

- 1) The significant contributions which changed the scenario of science and technology in the field of Mathematics and Astronomy as a whole were- concept of zero, decimal system, numeral notations, binary numbers, Chakravala algorithms, knowledge of measurements, theory of atom,

heliocentric theory, Kharosthi numerals, arithmetic, algebra, geometry and astronomy.

**Democracy, Civil
Society and
Ecology**

Check your progress: 2

- 1) The technological impacts of industrial revolution included:
 - Use of new materials
 - Use of new energy sources
 - Discovery of innovative machineries
 - Introduction of factory system and division of labor
 - Development in transportation and communication
 - Increased application of science in industry

Check your progress: 3

- 1) By adopting new technology, the Third World nations will have the chance to have a better link with rest of the world. During the last two decades, there has been a major technological innovation in the field of electronics and informatics, biotechnology, genetic engineering, lasers and remote sensing, virtual payment systems, virtual shopping, virtual home assistance and self-driving cars.

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UNIT 5 ENVIRONMENTAL ETHICS

Structure

- 5.0 Introduction
- 5.1 Learning Outcomes
- 5.2 Definition of Environmental Ethics
- 5.3 Approaches of Environmental Ethics
- 5.4 Environmental Attitudes
- 5.5 Environmental Justice
 - 5.5.1 Importance of Environmental Justice
- 5.6 Human Rights and Environment
- 5.7 Population, Consumption and Future Generations
 - 5.7.1 IPAT Equation
- 5.8 Indigenous Knowledge and Its Importance
- 5.9 Let Us Sum Up
- 5.10 Key Words
- 5.11 Further Readings
- 5.12 Check Your Progress: Possible Answers

5.0 INTRODUCTION

Man has always tried to build a mutual relationship with nature in spite of the fact during the course of evolution, the environment has been altered. Human beings have satisfied their needs taking the resources from the planet and altogether making attempts to renew those resources. Today also we need to understand the value of precious natural resources and unique nature's services and functions. Therefore it is our responsibility to see that the planet remains habitable not only for us but for the other living organisms as well. This notion has given rise to the concept of environmental ethics. It is the moral responsibility of the individuals to act according to his knowledge and capabilities for the welfare of the society and communities. Therefore ethics play an important role in our society. Individuals who are committed to protect the earth are making significant changes in their lifestyles so as to reduce their personal impact. In this unit you will learn more about environmental ethics and several principles pertaining to it. The importance of indigenous knowledge and essentials of development practices are also discussed in this unit.

5.1 LEARNING OUTCOMES

After reading this unit, you will be able to

- define environmental ethics;
- explain the principles of environmental ethics;
- justify safe environment as a human right; and

- describe the importance of indigenous knowledge in conserving nature.

5.2 DEFINITION OF ENVIRONMENTAL ETHICS

Ethics is a division of philosophy that determines what is right and what is wrong. In the same way environmental ethics help us to understand the association between humans and nature. Environmental issues must consider individual and societal ethics. Environmental ethics fosters considering ethics and codes of conduct to the nature and the relationship that exists between humans and other forms of life on this earth. Ethical issues regarding environment were discussed earlier also but environmental ethics as a separate branch emerged when global conferences and events concerning environment took place during 1970s. International events such as Stockholm Declaration and Brundtland commission established the importance of environment and any activity which harmed the components of nature was criticised. Environmental ethics came to the fore at this time when there was increased awareness about the impact of population on world's environment. According to Aldo Leopold, 1949 "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong when it tends otherwise..." Environmental ethics is the idealistic situation which demonstrates the ethical relationship between human beings and environment. In other words it explains the appropriate behaviour of human beings while utilizing the resources and services of nature.

It helps define man's moral and ethical obligations towards the environment. Human values are considered to be an important factor when looking at environmental ethics. In other words, it means that humans give importance to certain functions and on the basis of the relative importance of a particular function decide whether to exploit or conserve a particular resource. Human values differ from person to person as every individual does not accord the same value on nature's components. For example, for a poor person cutting down the forest is necessary to clear the land for agriculture whereas another person who lives in city may consider it unethical because he knows that forests destruction has lot many environmental consequences like increase in carbon- di-oxide. Thus in this sense environmental ethics are quite relative and vary from person to person and situation to situation. There are several arguments and counter arguments on the justification of man's interaction with environment.

For example: Is it ethically correct to build a hydroelectric dam inspite of the fact that it will cause displacement of thousand so people or will disrupt the migration pattern of some fishes.

Environmental ethics pertains to our commitment and responsibilities towards environment. It directs us to take care of our surroundings. Environmental ethics strives to bring together human values, moral principles, and optimum decision making in the overall interest of the environment and human community. The term was used in US for the first time in 1970. According to Wikipedia "Environmental ethics is the part of environmental philosophy which considers extending the traditional boundaries of ethics from solely including humans to including the non-

human world. It exerts influence on a large range of disciplines including environmental law, environmental sociology, ecotheology, ecological economics, ecology and environmental geography.”

All the above definitions emphasize equal responsibilities to establish equality. Provision of food and other materials is necessary but satisfaction of aesthetic human needs is also equally important. However, over exploitation of resources by growing human population has upset the natural balance and causes ecological problems.

Climate Justice

The term climate justice denotes a term to consider global warming as an ethical and political issue, rather than one treating it as environmental or physical issue. We have been discussing climate change as an environmental issue with focus on emissions, rising temperatures and melting ice and so on so forth but in all these discussions we missed out a very important component i.e. the people. Climate justice thus recognises the human dimension of climate change and helps us to understand the challenges faced by the communities most vulnerable to climate change. The people who are affected are generally the ones who contribute least towards climate change. This injustice can be corrected by taking suitable measures like achieving zero carbon economy and providing equal rights of development to all.

5.3 APPROACHES OF ENVIRONMENTAL ETHICS

Whenever you are debating on environmental issue you must have come across varied opinions. There are always arguments and counter arguments on the cause and effects of environmental problems. It is so because individuals differ on their obligations concerning the environment. Some value human concerns more than nature and justifies the exploitation of environment for the benefit of human beings. This view is known as anthropocentric view. A person with this viewpoint considers nature as a resource which is under his own disposal. Bentham, the founder of utilitarianism, put forward the viewpoint that the ultimate goal of ethics should be to cause the greatest good for the greatest number of people. This is also known as utilitarian justification because it stresses that conservation of nature is important as it provides individuals with direct economic benefits. For example, mangrove trees are the source of livelihood for the fishermen who live on fish, shrimp or other aquatic resources. The utilitarian justification for species preservation is based on recognizing the enormous benefits of biodiversity for human beings.

Ecological justification views the ecosystems as the source of the survival of certain species. For example, in case of degraded ecosystems, many species are on the verge of extinction like tigers or great Indian bustard. Same is the case of mangroves which provide habitat to the fishes. The ecological justification for species protection is based on understanding the roles of all the species in maintaining ecosystem functions. Although it is not possible to describe the importance of every species in maintaining the balance of ecosystem therefore we assume that each species is ecologically important

and work towards the conservation of the entire ecosystem.

There are three primary theories regarding environmental ethics. They are as follows:

Anthropocentrism: Anthropocentrism view opines that human comfort and happiness is the overall environmental responsibility. It argues that human beings are the most important entities of the universe and so the assumption here is that human beings can utilize and exploit the environment for their interests and making their lives better. Therefore, environment is crucial to their existence and hence it is the duty of human beings to protect the nature and must ensure the sustainability and longevity of human life on this planet. The anthropocentric view is considered to be responsible for causing global warming, pollution, water scarcity, loss of biodiversity and many other problems.

Biocentrism: Biocentrism believes that all forms of life are equal and according to Natural rights, they have an inherent right to exist. It is an ethical point of view and applies to all living things. Roots of biocentric ethics lies in our ancient culture and religions. Worshipping of animals and protection of trees has been extensively described in our traditions and historical accounts. However, there are numerous challenges in practising biocentrism. It is too demanding and even eating meat or even vegetables come under violation of this approach. For this, some propose that distinction between basic and non basic interests must be clearly made. For example, hunting of animals is unethical but killing animals in self defence is considered appropriate.

Ecocentrism: Another approach to environmental responsibility is called ecocentrism. It believes that environment must be protected considering moral aspects and not because human beings are deriving some profit out of it. It is a nature centred system of values unlike anthropocentrism which is basically human centred. It differs from biocentrism in the sense that the abiotic components of environment are also considered important under it. It believes environment as a whole and humans have moral responsibility of protecting it.

5.4 ENVIRONMENTAL ATTITUDES

Environmental attitudes determine our behaviour towards environment which affects its quality. The type of attitude we have forms our action towards preserving or destroying environment.

The three most common attitudes are:

Development ethic- This attitude is anthropocentric by nature. It believes in the supremacy of human beings over all other organisms. Therefore as per this attitude, Earth's resources are only for the benefit of human beings. Present day environmental degradation is the result of development approach and

Preservation ethic- This attitude is ecocentric by nature. It believes that nature and its resources are indispensable and should be protected at all cost.

Proponents of this attitude consider preservation of environment as most important.

Conservation ethic- This attitude seeks to find a balance between unplanned development and preservationist attitude. The proponents of this attitude promote human well being but also make efforts for environmental protection.

There are several factors which influence Environmental Attitudes such as demographic, social and psychological factors.

Sacred Groves

India has a long tradition of conservation of natural resources by associating local deities with them. Such areas which are preserved and where unlawful activities are restricted are known as sacred groves. The size of sacred groves varies from few trees to dense forests. Ancient communities used to live near forests and rivers. Hence were considered pious. Any harm to these resources was considered an act against God's wish. Cutting of trees and hunting of animals is strictly prohibited in these areas. By this biodiversity conservation was carried out. Today also, there are various places demarcated as sacred groves. It is considered as a biodiversity conservation measure. These are sustainable measures adopted by local communities. However, in today's times, the sacred groves are on decline due to deforestation and urbanization. Further, there is also a change in the social beliefs of the people.

5.5 ENVIRONMENTAL JUSTICE

You must not be aware that in Houston in 1970s, the landfills and incinerators were situated in communities of colour than their white neighbourhoods. This realization by Robert Bullard, a sociologist laid the foundation of the term called environmental justice. Later Robert Bullard was called as the father of environmental justice. 17 Principles of Environmental Justice were framed in 1991 and after that at the Earth Summit in 1992; Environmental justice was discussed at the international platform.

Environmental justice denotes fair and equal treatment to all human beings irrespective of their nationality, religion, sex, income or race in relation to execution of environmental regulations, laws and policies. It implies right to safe and secure environment to all human beings. You must have seen children, women and poor workers exposed to toxic chemicals and harmful gases in factories or forcibly consuming contaminated water. All these are examples of environmental injustice. People residing in hazard prone regions or near polluting industries are also the victims of environmental pollution.

Environmental justice is thus an important tool to provide clean and healthy living conditions to the masses. It becomes all the more important for the people who have been traditionally living near the polluted sites and have no access to clean water. For instance, one of the environmental justice issues is inadequate access to healthy food. Certain communities in society have no access, availability and affordability to healthy foods. Therefore, unbiased distribution of environmental benefits, participation of all the communities in

environmental decision-making, giving importance and recognizing the utility of local knowledge and right of individuals to live a decent life all come under the umbrella of environmental justice.

5.5.1 Importance of Environmental Justice

Environmental justice is important to provide the following basic functions:

Better Human Relationship With Environment – Our relationship with environment depends on what value we assign to its various parts and subparts. Justice towards nature shapes our thinking and attitude. It is necessary so that we are able to assign value to the nature's services and understand that it is vital for the healthy survival of living beings on this planet. Environmental pollution which threatens the life of living organisms must be condemned. When people understand that environmental justice protects them against nature's vagaries, they tend to protect the environment. Hence a congenial relationship develops between nature and human beings.

Natural Resource Conservation- Human beings help to conserve the natural resources when environmental justice is being followed. When people follow environmental justice, it makes them clear that earth's resources are free to be used and every individual should have equal access of using them. They realise that unfair distribution and usage of natural resources make them further scarce and limited. Unequal distribution of these resources may lead to resource conflicts and wars.

Ensuring Long Term Sustainability- The basic objective of all conservation measures is to ensure the sustainability of the resources so that the coming generations are also benefitted and are able to enjoy the benefits. Environmental justice strengthens the idea of inter generational and intra generational equity. It contributes in checking indefinite and uncontrolled extraction of natural resources. It strengthens the central idea of sustainability and reduces the dependence on non-renewable resources.

Equal Distribution of Resources – Environmental justice implies fair and equal distribution of resources which are provided to all by virtue of their existence as human beings. The resources are unfairly distributed on earth and therefore the rate of their exploitation also differs from region to region. By assigning environmental justice, all the basic resources such as water, soil, food are made available to all without any distinction. Pollution abatement measure comes in this category as they arrive to provide clean air to all the masses. Environmental justice thus stresses on equal distribution of resources.

Strengthening Environmental Laws, Policies and Regulations- Environmental justice supports the policies and regulations which govern the environment. It supports the principle that all people and communities are entitled to equal protection of our environmental laws. It treats all people equal while enforcing environmental regulations.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit

1. Which of the following is an example of environmental justice?

- a) Absence of fast food restaurants
- b) No farms near community
- c) No access to food
- d) No culinary schools in the community

2. Differentiate between biocentrism and ecocentrism.

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5.6 HUMAN RIGHTS AND ENVIRONMENT

Human rights are fundamental rights guaranteed to human beings because of their inherent nature. They are applicable to all human beings irrespective of their religion, place of origin, colour or sex. Right to food, right to education, right to live peacefully, all are examples of human rights. In today's era, when the environment is getting polluted, the need has emerged to declare right to safe environment as a human right. The fundamental right to life is dying out because we are increasingly exposed to harmful chemicals in air, water and even our food. We are forced to consume contaminated water, breathe polluted air and eat food burdened with pesticides. Environmental conditions indicate the extent to which people enjoy their basic human rights. A healthy environment provides healthy living conditions for organisms to thrive. Further, it is understood that human rights cannot be secured in a degraded or polluted environment. It is time to recognize that those who pollute or destroy the natural environment are not just committing a crime against nature, but are violating human rights as well.

Internationally much has been discussed on the linkages between human rights, health and environmental protection. To address the specific problems in these areas, legal instruments such as laws and rules, specialized departments and agencies have been set up. In the past, health, environment and human rights were dealt separately. The links between these were established at the Stockholm conference in 1972. At the Stockholm concluding session, the participants declared that Man is both creature and moulder of his environment, which gives him physical sustenance and affords him the opportunity for intellectual, moral, social and spiritual growth. Both aspects of man's environment, the natural and the man-made, are essential to his well-being and to the enjoyment of basic human rights even the right to life itself. Principle 1 of the *Stockholm Declaration*

established a foundation for linking human rights, health, and environmental protection, declaring that Man has the fundamental right to freedom, equality and adequate conditions of life, in an environment of a quality that permits a life of dignity and well-being.

5.7 POPULATION, CONSUMPTION AND FUTURE GENERATIONS

The world's population increased to six billion people during the 20th century and it may reach 9 billion by 2050. However, the per capita income is also increasing and people are consuming the earth's resources at an alarming rate. The rich countries of the world have the major share in consumption. For example, the rich countries of the world are responsible for most of the world's carbon dioxide emissions and people there use most of the world's resources and produce more waste.

In view of this, it is often discussed that increasing population and consumerism is going to deprive the future generations of the vital resources that are essential for survival. The quality and quantity of the resources is dwindling and we must understand that today's decisions and actions are going to affect the future of coming generations. There is a Native American saying that "We have not inherited the Earth from our fathers. We are borrowing it from our children". Therefore one must have concerns for the coming generations regarding resource availability. Every discovery and every advancement is our endowment to the coming generations. The knowledge and the experiences in the form of economic prosperity and technological progress are passed on from one generation to another. However, unplanned development and reckless exploitation of resources may produce more harm than good. The non renewable resources are scarce and limited. We must ensure that these are available for our coming generations. Protection of the earth is everyone's responsibility and our actions and connections will create a common future for all.

5.7.1 IPAT Equation

IPAT equation was proposed by Ehrlich and Holdren in the early 1970s. It conveys the idea that environmental impact (I) is the result of population (P), affluence (A) and technology (T). The equation was basically used to know the impact of human activities on environment.

Alarming increase in World population growth raises a question whether Earth would be able to support such huge numbers and whether enough food, water and energy is available for all. However, it is now clear that population alone is not responsible for environmental degradation but how people live also is responsible. It is consumption habits or the affluence of the people that influence the overall health of the environment. If a society becomes more affluent, there is increased consumption and use of natural resources. For example, people living in high income countries eat more meat-based diet, as opposed to people in low income countries who eat mostly grains.

Let us understand this by the following example:

Suppose a poor person A lives in a village where there is no electricity or vehicles, his ecological footprint, will be lower than person B who lives in a city and consumes electricity for carrying out day to day activities.

5.8 INDIGENOUS KNOWLEDGE AND ITS IMPORTANCE

The meaning and nature of traditional knowledge has been explained in the current module. The knowledge of the tribal communities about the medicinal properties of certain plants has been described. The module explains the importance of traditional knowledge in ensuring the optimum utilization of resources and promoting their sustainable use. It was discussed that in the move towards conservation of Traditional Knowledge and sustainable development, efforts should be made to capture, document and safeguard the ethnicity of the tribal communities.

Indigenous Knowledge (IK) is a local knowledge possessed by a culture or society which influences local level decision making in fields of agriculture, health care and natural resource management. The indigenous knowledge enables poor and marginalized group of people to have greater control over their resources.

Therefore we can say that indigenous knowledge is very functional and lively in nature. This knowledge forms an integral part of ancient folk lore which gets passed on from one generation to another by word of mouth and experiences. So, proper understanding of folk life can help us gain better insights about the indigenous knowledge at a particular place. Around 85 per cent of India's population lives in the villages and 600 million of them are involved in agriculture. There is a dire need to focus on their development and to sustain and enhance their capacity to produce food grains for the entire nation by carefully utilizing natural resources. The outstanding traditional knowledge they had inherited from their ancestors regarding plants, animals, birds, soil, water, flora and fauna should be leveraged for sustainable development. Documenting the best practices and disseminating them to other farmers across the country would result in many other farmers adopting productive methods.

India has a rich heritage of Indigenous knowledge/ traditional medicine from plants. The traditional knowledge is fading up due to generation gap trapping that knowledge is very crucial as well as important. Biodiversity, indigenous knowledge, and sustainable development are very closely linked.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. What are the major environmental problems of today that are threatening the full enjoyment of human rights?

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2 To what extent is the indigenous knowledge relevant in today's time?

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5.9 LET US SUM UP

Environmental ethics relates to our moral obligation towards the environment. There are several approaches to environmental ethics viz. anthropocentrism, biocentrism, and ecocentrism. Due to increasing apprehensions regarding environment and the growing awareness about the impact of human activities, there is a growing realisation that environmental ethics has enormous scope. The environment is not something which we can remove oneself from. Human beings have started realising that they have a right to healthy and safe environment. The unit explains the meaning of environmental ethic, environmental justice and the role of indigenous knowledge in protecting our ancient wisdom on environment. When a person realizes his environmental obligations, he is able to think how an action or an event affects the entire range of environmental settings. For example, climate change which affects the world in various different ways. Further, the lifestyle of a person also impacts the environment and the future generations must be taken care of while consuming the limited resources present on the earth. The unit explains that it is not just the numbers but the ways of living of population which has a role in deteriorating the environment of a particular place.

5.10 KEY WORDS

Environmental Justice: It is the fair treatment and meaningful involvement of all people regardless of race, colour, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies.

Ethics Ethics: Moral principles that govern a person's behaviour or the conducting of an activity.

Human Rights: Human rights are moral principles or norms that describe certain standards of human behaviour, and are regularly protected as legal rights in municipal and international law.

Indigenous Knowledge: It refers to knowledge systems embedded in the cultural traditions of regional, indigenous, or local communities.

5.11 FURTHER READINGS

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5.12 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) Your answers must include the following points:
 - c
- 2) Your answers must include the following points:
 - Difference in focus and values.
 - Examples

Check Your Progress: 2

- 1) Your answers must include the following points:
 - Air and water polluted
 - Congestion on roads and streets
 - No aesthetic value remaining
 - No space to live
- 2) Your answers must include the following points:
 - The knowledge is unique and eco-friendly in nature
 - Medicinal value of plants needs to be recognized
 - Traditional water conservation methods are still relevant.